

A MERGED CONTINUATION OF THE SUPPLIED SURCOMPLEX MANUSCRIPTS

Surcomplex Polynomial Algebra

Root Geometry, Newton Profiles,
Multiscale Stability, and Residue Duality

September 21, 2026

Abstract

We develop finite-degree polynomial algebra over the surcomplex class field $\mathbf{SC} = \mathbf{No}[i]$, keeping two geometries deliberately apart: the surreal-valued modulus, which gives ordered disks, and the natural Hahn valuation, which gives valuation balls of arbitrary ordered-group rank. All proofs take place in set-sized algebraically closed Hahn workspaces $K_\Gamma = \mathbb{C}((t^\Gamma))$, and every degree is an ordinary finite integer.

The ordered part proves root bounds at arbitrary surreal radii, Gauss–Lucas with surreal convex weights and its weighted incomplete-polynomial converse, Jensen’s disk theorem, a sharp Schoenberg second-moment inequality with its collinearity equality case, a precisely delimited real-closed-field transfer giving a polynomial Rouché theorem on order circles, and Hermite’s signature criterion for counting real roots. It also determines the *exact* root locus of a polynomial under bounded coefficient uncertainty, in both geometries, and exhibits the structural contrast between them: a sum of available magnitudes against a minimum of available valuations.

The scale-sensitive part is organized by one initial-polynomial identity. It counts roots in every closed ball, open ball, shell and residue direction, proves Newton’s root-valuation rule over an arbitrary ordered exponent group, gives exact polynomial image balls and fiber degrees, and — after differentiation — shows that a ball containing $k \geq 1$ roots contains exactly $k - r$ roots of the r th derivative for $1 \leq r \leq k$. A finite root-cluster tree, an explicit branch polynomial, two discriminant layer formulas and a nearest neighbour identity locate the intervening critical directions.

Three perturbation theorems with genuinely different hypotheses are kept apart: an optimal ε/n valuation-Hölder matching that needs no root separation at all; a rootwise threshold $T(P) = \max_i(d_i + \delta_i)$ that preserves the entire matrix of pairwise distance valuations together with their leading coefficients, the derivative valuations and the discriminant valuation, with an exact displacement valuation and a second-order Newton error term; and the coarser discriminant certificate, shown to demand strictly more precision on an explicit quartic and to be sharp at equality on an explicit quadratic.

Support-controlled coprime factor lifting is proved with the certificate S^* , and in a parameter form over $\mathcal{H}_\Gamma(U) = \mathcal{O}(U)((t^\Gamma))$ with no shrinking of the ordinary domain. The algebraic package ends with a universal residue pairing that is perfect over every commutative ring and through every collision, an explicit dual basis with no discriminant denominator, the trace identity $\mathrm{Tr}(m_h) = \lambda_P(P'h)$, finite polynomial maps with branch-value and ramification formulas, a class-safe Nullstellensatz for finite data, and — in several variables — finite free normal forms, a coefficient-independent Bézoutian pairing, the Jacobian trace formula through collisions, and an identification of the supplied coherent contour residue in a delimited polynomial subclass.

Status. This article is a merged, theorem-driven research exposition. It does not claim that any statement below is a first occurrence in the literature, and it does not claim to resolve a named open conjecture. Several central statements are classical over real-closed, valued or arbitrary commutative-ring settings, and are identified as such. The literature search behind the attributions was targeted rather than exhaustive. The article has not been independently refereed. The repository’s formalization ledger records partial Lean coverage; no complete formalization of this article is claimed. The symbolic checks recorded in the source archives validate displayed finite examples and are not machine verification of the general proofs. Per-theorem non-claims are kept where they were made and are collected in [section 20](#).

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1 Subject, sources, and conventions

1.1 What changes when the coefficients are surcomplex

A surcomplex number is $x + iy$ with $x, y \in \mathbf{No}$ and $i^2 = -1$. That this class field is algebraically closed is established surreal theory [7, 8, 10], not a new fundamental theorem of algebra. For a polynomial of degree n , algebraic closedness already supplies n roots with multiplicity. It answers none of the geometric questions. How do roots sit relative to a disk of infinite radius? How many roots occupy a prescribed infinitesimal subcluster? Which critical points sit *between* two such subclusters? How much coefficient precision guarantees a one-to-one matching of roots, and when can an arbitrarily small perturbation make distinct roots collide? What survives a collision on the residue-theoretic side?

Two different structures answer these questions, and they must not be identified. The modulus $|x + iy| = \sqrt{x^2 + y^2}$ is surreal-valued and satisfies the ordinary triangle inequality. The natural valuation records a leading Hahn exponent and is ultrametric. An order disk $|z - a| \leq r$ is not the valuation ball $v(z - a) \geq v(r)$: for instance $B_{\geq}(0, 0)$ contains every finite element, not merely the modulus unit disk, while $2t^\rho \in B_{\geq}(0, \rho)$ but $2t^\rho \notin \{|z| \leq t^\rho\}$. The whole article is organized by that separation.

1.2 The three merged manuscripts

This article merges three supplied manuscripts that share a title and a spine. We cite them as [1, 2, 3] and identify their individual contributions wherever they differ.

[2] (“Order Geometry, Newton Profiles, and Scale-Resolved Stability”) is the largest and supplies the backbone of the scale-sensitive theory: the initial-polynomial identity (theorem 8.2), Newton’s root-valuation rule over an arbitrary ordered exponent group (theorem 9.1), the exact image balls and fiber degrees (theorem 10.1), the r th-derivative critical count (theorem 11.2), the optimal ε/n valuation-Hölder matching (theorem 13.1), the second-order Newton error estimate (theorem 13.3(13.9)), the Jensen disk theorem (theorem 4.4) and the sharp Schoenberg second-moment inequality with its equality case (theorem 5.3).

[1] (“Factorization, Root Geometry, Multiscale Stability, and Residue Duality”) supplies the exact coefficient-uncertainty root loci in both geometries (theorem 7.1), the named rootwise threshold $T(P)$ with its full cluster-preservation conclusion and its two-sided calibration (theorem 13.3, corollary 13.4, and examples 13.5 and 13.6), the telescoping cluster formula for the discriminant valuation (proposition 11.9), the universal residue pairing with an explicit dual basis and no discriminant denominator (theorem 14.1), the universal trace identity (theorem 14.2), the finite flat map, branch-value and ramification package (theorems 10.3 and 10.4 and proposition 10.5), the class-safe Nullstellensatz for finite data (theorem 15.2) and the non-Noetherian witness in remark 15.3.

[3] (“Ordered Geometry, Valuatative Root Trees, and Residue Duality”) is the only one of the three to carry the degree-controlled complete-intersection and Bézoutian package beyond one variable — all three leave one variable somewhere, the other two for the Nullstellensatz and finite-algebra material. It supplies Hermite’s signature criterion over surreal real closed fields (theorem 6.4), the weighted incomplete-polynomial converse to Gauss–Lucas (proposition 4.3), the nearest-neighbour critical-point identity (corollary 11.5), the branch polynomial at a node of the cluster tree and the tree form of the discriminant (theorem 11.6 and proposition 11.10), the escape example (example 2.8), and the entire several-variable package: finite free normal forms over an arbitrary commutative

ring (theorem 16.1), the coefficient-independent Bézoutian pairing (theorem 17.1), the Bézoutian kernel and Jacobian trace identity (theorem 17.2), Euler–Jacobi vanishing (corollary 17.3), global polynomial families on one ordinary parameter domain (theorem 17.4) and the identification of the supplied coherent contour residue in a delimited polynomial subclass (theorem 17.5).

All three are explicitly continuations of earlier supplied manuscripts, cited here as [4] (*Surcomplex Analysis: Infinitesimal Calculus, Hahn-Coherent Holomorphy, and Contour Theory*), [5] (*Hartogs Coherence, Finite Maps, and Residue Duality over the Surcomplex Numbers*) and [6] (*Global Divisors and Cohomological Obstructions in Hahn-Coherent Surcomplex Analysis*). We take from them the notation $t^\gamma = \omega^{-\gamma}$, set-sized Hahn workspaces, strong summability, the coherent category $\mathcal{H}_\Gamma(U)$ and its strong evaluation, and nothing else. As all three merged manuscripts say in one form or another: we use that framework as motivation, not as a prerequisite for the polynomial results, and we do not silently identify their coefficient sheaves with ordinary convergent power-series rings or assume their proposed novelty assessments as facts.

1.3 Conventions: one symbol per concept

The three sources use three notations for several of the same objects. We fix one symbol per concept here and use only that symbol thereafter. Synonyms are recorded once, in this paragraph, and never used again.

- **Workspace.** $\Gamma \subseteq \mathbf{No}$ is a set-sized divisible ordered additive subgroup; $F = F_\Gamma = \mathbb{R}((t^\Gamma))$ and $K = K_\Gamma = \mathbb{C}((t^\Gamma)) = F_\Gamma[i]$, with $t^\gamma = \omega^{-\gamma}$. (Synonym in the sources: R_Γ for F_Γ .)
- **Valuation and reduction.** v , lc , $\mathcal{O}_K = \{v \geq 0\}$, $\mathfrak{m}_K = \{v > 0\}$, and the coefficient-at-zero map $\text{st} : \mathcal{O}_K \rightarrow \mathbb{C}$. The map st is called the *standard part*; *reduction* and *shadow* are synonyms used in the sources, and red is the symbol one of them uses for it. We write st throughout and never red . The topology it induces is called the *standard-part topology*; the names *residue topology*, *reduction topology* and *shadow topology* occur in the sources for the same thing and are not used again.
- **Balls and disks.** $B_{\geq}(a, \rho) = \{v(z - a) \geq \rho\}$, $B_{>}(a, \rho) = \{v(z - a) > \rho\}$ are the closed and open *valuation balls*; $D(a, r)$, $\overline{D}(a, r)$ are the open and closed *order disks*. (Synonyms: B_{cl} , B_{op} ; “valuative disk”.) The words “open” and “closed” describe the defining inequalities only; no fine-topological connectedness or compactness is used anywhere.
- **Scale data.** The scale exponent is always $\rho \in \Gamma$ (synonyms in the sources: δ , γ). The weighted Gauss valuation is $w_{a,\rho}(P)$ (synonym: $g_P(a, \rho)$) and the *initial polynomial* is $I_{a,\rho}(P) \in \mathbb{C}[X]$ (synonym: the *reduction polynomial* $R_{a,\rho}$). Its variable is always X ; the polynomial variable over K is always z .
- **Roots and condition numbers.** Roots are α_i (synonym: r_i); $d_{ij} = v(\alpha_i - \alpha_j)$; $d_i = v(P'(\alpha_i)) = \sum_{j \neq i} d_{ij}$ (synonym: σ_i); $\delta_i = \max_{j \neq i} d_{ij}$; and the threshold is

$$T(P) = \max_i (d_i + \delta_i)$$

(synonym: $\tau(P)$). A perturbation is always E with coefficient precision $\varepsilon = \min_j v([z^j]E)$ (synonyms: H with μ ; Δ).

- **Residues.** $\lambda_P(h) = [z^{n-1}] \text{rem}_P h$ in one variable and $\lambda_F(H) = [x^{\text{top}}] \text{NF}_F(H)$ in several; Δ_P , Δ_F are Bézoutians; J_F is the Jacobian determinant.

1.4 Ring discipline

Four coefficient rings with nearly identical notation circulate in the wider set of manuscripts from which this report is drawn: a fixed-domain ring, a common-domain germ ring, a radius-free ring whose Hahn coefficients are individually convergent germs with no common polydisk, and a formal-coefficient ring. Theorems about them are *not* interchangeable, and the notation does not distinguish them. We therefore state the rule once and then obey it.

Ring	Role in this article
$K = \mathbb{C}((t^\Gamma))$, and $\mathcal{O}_K, \mathfrak{m}_K$	The coefficient field of every univariate polynomial theorem below, and its valuation ring. Everything in sections 2 to 15 that mentions roots, balls, valuations or precision is stated over K or $\mathcal{O}_K$.
$\mathcal{H}_\Gamma(U) = \mathcal{O}(U)((t^\Gamma))$, its nonnegative-support subring $\mathcal{H}_{\Gamma, \geq 0}(U)$, and the positive-support ideal $\mathcal{H}_{\Gamma, > 0}(U) \subseteq \mathcal{H}_{\Gamma, \geq 0}(U)$	The <i>fixed ordinary domain</i> parameter ring: $U \subseteq \mathbb{C}^d$ is one ordinary connected domain, fixed once and never shrunk. Exactly three theorems are stated over it: theorems 12.5, 17.4 and 17.5 , together with section 14.4 .
An arbitrary commutative ring R (one variable) or A (several) with identity $\mathbb{C}[z]((t^\Gamma))$	The coefficient ring of the universal duality theorems theorems 14.1, 14.2, 16.1, 17.1 and 17.2 . No Noetherian, reduced, domain or field hypothesis is used there. Appears only as a <i>counterexample ring</i> in remark 2.7 and section 18 : it permits unbounded polynomial degrees among Hahn coefficients and is never a coefficient ring for a theorem.

Every theorem below names its ring in its statement. No theorem is transferred from one of these rings to another, and in particular:

- nothing below is stated over a radius-free ring $\mathbb{C}\{z\}((t^\Gamma))$ whose Hahn coefficients are individually convergent germs with no common polydisk, and nothing below is stated over a formal-coefficient ring $\mathbb{C}[[z]]((t^\Gamma))$. Neither ring occurs in any of the three merged manuscripts, and no result here should be read as a statement about them;
- the parameter theorems are stated for one *fixed* ordinary domain U . They are not germ statements, and their proofs give no common radius and claim none;
- the universal-ring duality theorems are stated over an arbitrary commutative ring. Specializing them to $\mathcal{H}_\Gamma(U)$ or to K is a ring homomorphism and is always performed explicitly, never assumed.

1.5 Degree, and what “polynomial” means

Throughout, a polynomial has *ordinary finite degree* in its variables. Its coefficients may have arbitrary set-sized Hahn supports. We do not use ordinal degrees, proper-class sums, or a polynomial-like series of unbounded degree under the name “polynomial.” Every theorem about an individual finite root configuration is proved in a set-sized field containing its data and then interpreted in **SC**. The degree hypothesis is a genuine restriction, not a disguised assertion about all isolated complete intersections; [section 18](#) makes the dividing line exact.

2 Foundations: workspaces, supports, and two notions of size

2.1 A set-sized field for every finite problem

We use a class set theory admitting the usual treatment of **No**, for example NBG with global choice, in the same way as the supplied manuscripts. Each individual support, recursion, coefficient list, matrix and algebraic certificate is a *set*. Conway normal forms give, for a set-sized ordered abelian subgroup $\Gamma \subseteq \mathbf{No}$,

$$K_\Gamma = \mathbb{C}((t^\Gamma)) \subseteq \mathbf{SC}, \quad F_\Gamma = \mathbb{R}((t^\Gamma)), \quad t^\gamma = \omega^{-\gamma}, \quad (2.1)$$

an element being $\sum_{\gamma \in S} c_\gamma t^\gamma$ with $S \subseteq \Gamma$ well ordered in the increasing order and $c_\gamma \in \mathbb{C}$.

Proposition 2.1 (A workspace for every set of data). *Every set of surcomplex constants is contained in a field K_Γ with Γ divisible and set-sized. Such a field is algebraically closed; its fixed field under complex conjugation is the real closed ordered field $F_\Gamma = \mathbb{R}((t^\Gamma))$, and $K_\Gamma = F_\Gamma[i]$. A polynomial already split in K_Γ acquires no additional roots, and no changed multiplicities, after passage to a larger surcomplex workspace. More generally, if $A = K_\Gamma[x_1, \dots, x_s]/I$ is finite-dimensional, every surcomplex point annihilating I has all coordinates in K_Γ , and its local multiplicity is unchanged by enlargement.*

Proof. Take the union of the normal-form supports of the given constants, generate its additive subgroup in **No**, and take the divisible hull; these are set operations. The Hahn-field algebraic-closure theorem applies because Γ is divisible and the residue field $\mathbb{C}$ is algebraically closed [10, Corollary 4]. Real and imaginary parts of a Hahn series lie in F_Γ , so $K_\Gamma = F_\Gamma[i]$; an ordered field whose extension by i is algebraically closed is real closed, which gives the second assertion. If $P = a_n \prod_{j=1}^n (z - \alpha_j)$ already in $K_\Gamma[z]$, that identity persists in every extension field, and a product of linear factors cannot vanish at an element different from all the α_j . For the last assertion, the coordinates of a point of a finite-dimensional quotient are roots of the characteristic polynomials of the commuting multiplication matrices, hence lie in K_Γ ; after scalar extension each local factor still has a nilpotent maximal ideal and the extension field as residue field, so it is still local of the same vector-space dimension. $\square$

We write F, K for one chosen pair and enlarge workspaces to include centers, scales and finitely many parameters without further comment. A theorem stated for all surcomplex centers and scales means that each particular finite selection is placed in such a workspace. It does not require one set-sized field to contain every surreal number.

Remark 2.2 (What **SC**[z] is not). The notation **SC**[z] is convenient for the class of finite polynomials, but it does not authorize an application of Zorn's lemma to a proper-class family of ideals. Euclidean calculations concern finitely many polynomials. Maximal ideals, finite-dimensional algebras and the Nullstellensatz of [section 15](#) are first handled in the *set* ring $K_\Gamma[z]$ or $K_\Gamma[x_1, \dots, x_d]$, and only then interpreted in **SC**.

2.2 Modulus, valuation, and the standard part

For $z = x + iy \in K$ put $\bar{z} = x - iy$ and $|z| = \sqrt{z\bar{z}} = \sqrt{x^2 + y^2} \in F_{>0}$, the positive square root existing by real closedness. The algebraic Cauchy–Schwarz identity $(x^2 + y^2)(u^2 + v^2) - (xu + yv)^2 = (xv - yu)^2$ gives $|zw| = |z||w|$ and $|z + w| \leq |z| + |w|$ after squaring two nonnegative sides. These arguments use no limits.

For $A \neq 0$ set $v(A) = \min \text{supp}(A)$, $\text{lc}(A) = A_{v(A)}$, and $v(0) = +\infty$. Then

$$v(AB) = v(A) + v(B), \quad v(A + B) \geq \min\{v(A), v(B)\}, \quad v(|A|) = v(A), \quad (2.2)$$

the last from the leading positive coefficient of $A\bar{A}$. Every nonzero ordinary complex number has valuation zero; in particular

$$v(m) = 0 \quad \text{for every nonzero integer } m. \quad (2.3)$$

This residue-characteristic-zero fact is responsible for the exact critical point counts of [section 11](#). Write $\mathcal{O}_K = \{v \geq 0\}$ and $\mathfrak{m}_K = \{v > 0\}$ for the finite elements and the infinitesimals; the coefficient at exponent zero gives $\text{st} : \mathcal{O}_K \rightarrow \mathbb{C}$ with $\mathcal{O}_K/\mathfrak{m}_K \cong \mathbb{C}$, and every finite A is uniquely $\text{st}(A) + \varepsilon$ with $\varepsilon \in \mathfrak{m}_K$.

Definition 2.3 (Order disks and valuation balls). For $a \in K$, $r \in F_{>0}$ and $\rho \in \Gamma$,

$$D(a, r) = \{z : |z - a| < r\}, \quad \bar{D}(a, r) = \{z : |z - a| \leq r\},$$

$$B_{\geq}(a, \rho) = \{z : v(z - a) \geq \rho\} = a + t^\rho \mathcal{O}_K, \quad B_{>}(a, \rho) = \{z : v(z - a) > \rho\} = a + t^\rho \mathfrak{m}_K.$$

A *residue-direction ball* at this center and scale is $a + t^\rho(c + \mathfrak{m}_K)$ for $c \in \mathbb{C}$; it is the open valuation ball $B_{>}(a + t^\rho c, \rho)$.

For example $2t^\rho \in B_{\geq}(0, \rho)$ but $2t^\rho \notin \bar{D}(0, t^\rho)$, and $|1 + 1| = 2$, so the modulus is not ultrametric. If $v(x) > v(y)$ with $y \neq 0$, then x/y is infinitesimal, so $|x| < |y|/m$ for every positive integer m : valuation geometry forgets ordinary nonzero complex factors while recording every surreal scale. This is exactly why its perturbation criteria differ from modulus criteria, a point made precise in [theorem 7.1](#).

2.3 The support lemma, and the two things it does not say

A family of Hahn series is *strongly summable* if the union of its supports is well ordered and only finitely many members contribute at each exponent; the sum is then defined by finite coefficientwise addition. For a well-ordered $S \subseteq \Gamma_{>0}$ put $S^* = \{e_1 + \cdots + e_k : k \geq 0, e_j \in S\}$, including the empty sum 0.

Lemma 2.4 (Hahn–Neumann support calculus). *If S, T are well-ordered subsets of an ordered abelian group, then $S + T$ is well ordered and each element has finitely many representations $s + t$. If S is well ordered and positive, then S^* is well ordered and each exponent has finitely many representations by nondecreasing finite words in S . Consequently, if T is well ordered, a family of monomial terms indexed by $b \in T$ and finite words $(e_1, \dots, e_k)$ in S , each supported at $b + e_1 + \cdots + e_k$, is strongly summable provided each such pair indexes only finitely many terms.*

Proof. The first two assertions are the classical support facts [9]; the same formulation is used in all three merged manuscripts. For the last, decompose an output exponent as $s + \eta$ with $\eta \in S^*$: there are finitely many such pairs, finitely many nondecreasing words for each η , and finitely many orderings of each finite word. Both requirements for strong summability have then been checked. $\square$

The lemma justifies $(1 + B)^{-1} = \sum_{k \geq 0} (-B)^k$ for B of positive support, as an *exact* Hahn identity. Two negative companions are load-bearing and are stated here once, since every construction below is checked coefficient by coefficient rather than by an improving estimate.

Remark 2.5 (Strong summability is not a valuation limit). [Lemma 2.4](#) does *not* say that the partial sums converge in the fine surreal topology. With $\Gamma = \mathbb{Q} + \mathbb{Q}\omega$ the strong sum of t^k over $k \geq 0$ is $(1 - t)^{-1}$, yet every remainder after finitely many terms exceeds the positive tolerance t^ω : the tails never reach valuation ω . Nor does the lemma say that $k\rho$ eventually exceeds every positive value-group element; that is false in higher rank. No argument of the form “the valuation improves by a fixed positive amount at each iteration” justifies any construction in this article, and we never substitute either false cofinality claim for [lemma 2.4](#).

Remark 2.6 (Finitely many dependencies is not finitely many exponents below). “Only finitely many members contribute at a given exponent” does not mean “only finitely many exponents lie below it.” The latter already fails below ω in $\mathbb{Q} + \mathbb{Q}\omega$, and it fails in [example 9.4](#) within a support of order type ω all of whose exponents are below 1. The distinction is what makes the finite-parameter-replacement steps in the recursions of [section 12](#) legitimate. Well-ordering also cannot be weakened to positivity: $\{1/m : m \geq 1\}$ is positive and not well ordered.

2.4 The coherent parameter ring on a fixed ordinary domain

For an ordinary connected domain $U \subseteq \mathbb{C}^d$, the supplied category is

$$\mathcal{H}_\Gamma(U) = \mathcal{O}(U)((t^\Gamma)),$$

Hahn families of *ordinary holomorphic* functions on *one common, fixed* ordinary domain U . The nonnegative-support series form the subring $\mathcal{H}_{\Gamma, \geq 0}(U)$; those of strictly positive support form its ideal $\mathcal{H}_{\Gamma, > 0}(U)$, which does not contain 1. A datum $F = \sum f_\gamma t^\gamma$ is evaluated at $c + \varepsilon$ with $c \in U$ and $\varepsilon \in \mathfrak{m}_K^d$ by Taylor expanding each ordinary coefficient,

$$F^\sharp(c + \varepsilon) = \sum_\gamma \sum_{\alpha \in \mathbb{N}^d} \frac{\partial^\alpha f_\gamma(c)}{\alpha!} t^\gamma \varepsilon^\alpha, \quad (2.4)$$

whose support lies in $\text{supp}(F) + (\bigcup_j \text{supp } \varepsilon_j)^*$ and has finite multiplicity by [lemma 2.4](#). This convention is inherited from [\[4, 5\]](#); it is not a new interpretation of a fine limit. The subscripts ≥ 0 and > 0 are used here with their stated meaning throughout, to avoid the notational mismatch between the supplied papers, where a superscript or subscript $+$ carries different conventions.

Remark 2.7 (Uniform degree, and a ring that lacks it). A polynomial with coefficients in $\mathcal{H}_\Gamma(U)$ has a uniform finite degree bound in z ; conversely, if every coefficient polynomial in a Hahn datum has degree at most one fixed finite n , collecting the $n + 1$ powers of z gives such a polynomial. Without that bound this fails. The ring $\mathbb{C}[z]((t^\Gamma))$ permits unbounded polynomial degrees among its Hahn coefficients, and

$$\sum_{k \geq 0} t^k z^k \quad (2.5)$$

belongs to it when $1 \in \Gamma$. It is a legitimate coherent datum on the thickening of every ordinary domain, where it equals $(1 - tz)^{-1}$; it is *not* a finite polynomial, and it is not a global pole-free function of all surreal scales, since the rational function has a pole at the infinite point $z = t^{-1}$. This elementary example separates the subject of this article from unrestricted Hahn-coherent entire data. All global polynomial evaluations here are finite sums.

Example 2.8 (A global root invisible in a finite monad). Let $t = \omega^{-1}$ and $F(z) = z + tz^2 = z(1 + tz)$. As a global polynomial it has the two roots 0 and $-t^{-1}$, and $K[z]/(F)$ has dimension two. On the thickening of any fixed ordinary disk about zero, $1 + tz$ is a Hahn-coherent unit with strong

inverse $\sum_{k \geq 0} (-tz)^k$, so the corresponding analytic quotient has rank one and sees only the root zero. Neither statement contradicts the other. In particular, positive Hahn support of a perturbation does *not* by itself preserve the global polynomial degree or the total number of roots over all of **SC**. The same phenomenon in the form $z^2 + tz^3 = z^2(1 + tz)$ shows a double root in the infinitesimal monad of zero together with a further root at $-t^{-1}$: the local standard part is z^2 while the global degree is three.

3 Finite polynomial algebra and multiplicity

Throughout this section K is an algebraically closed workspace as in [proposition 2.1](#), so all statements apply to surcomplex polynomials.

3.1 Factorization, division, and the squarefree part

Theorem 3.1 (Finite polynomial algebra over K). *Every nonconstant $P \in \mathbf{SC}[z]$ of degree n admits a factorization*

$$P(z) = a_n \prod_{j=1}^n (z - \alpha_j) = a_n \prod_{i=1}^s (z - \alpha_i)^{m_i}, \quad a_n \neq 0, \quad \sum_i m_i = n, \quad (3.1)$$

with the multiset of roots unique, all of them lying in one algebraically closed set-sized Hahn workspace containing the coefficients. For $A, B \in K[z]$ with $B \neq 0$ there are unique Q, R with $A = BQ + R$ and $\deg R < \deg B$; every pair not identically zero has a monic greatest common divisor $D = UA + VB$; and every ideal of $K[z]$ is principal. The multiplicity of α in P is the least j with $P^{(j)}(\alpha) \neq 0$, and

$$\frac{P'(z)}{P(z)} = \sum_i \frac{m_i}{z - \alpha_i}, \quad \gcd(P, P') = \prod_i (z - \alpha_i)^{m_i - 1} \quad (3.2)$$

for monic P , so P is squarefree exactly when $\gcd(P, P') = 1$.

Proof. Use [proposition 2.1](#), choose one root and divide by its linear factor; induction on the finite degree gives the factorization, and canceling a vanishing factor gives multiset uniqueness. Leading-term division lowers a nonnegative integer degree at each step, so it terminates and gives Q, R ; the Euclidean algorithm then produces D and the Bézout coefficients. A nonzero member of least degree generates any nonzero ideal; the zero ideal is generated by zero. The multiplicity characterization is the finite Taylor identity $P(\alpha + Y) = \sum_{j \leq n} (P^{(j)}(\alpha)/j!) Y^j$ together with characteristic zero. The product rule gives the first identity in (3.2); writing $P = (z - \alpha_i)^{m_i} U$ with $U(\alpha_i) \neq 0$ gives $\text{ord}_{\alpha_i} P' = m_i - 1$, because $m_i \neq 0$ in characteristic zero, which gives the second. $\square$

This is an application of established algebraic closedness, not a new proof of the surreal normal-form theorem. These are finite algebraic facts, unaffected by the sizes of the coefficients.

A monic polynomial in $\mathcal{O}_K[z]$ has all its roots in $\mathcal{O}_K$: at a root of negative valuation the leading term has strictly smaller valuation than every other term and cannot be cancelled. Conversely a monic polynomial with finite roots has finite coefficients, by Vieta. Reduction therefore gives the exact multiset identity

$$\text{st}(P)(z) = \prod_{j=1}^n (z - \text{st}(\alpha_j)) \quad (P \in \mathcal{O}_K[z] \text{ monic}). \quad (3.3)$$

Multiple roots of the reduction can split at many different infinitesimal scales; reduction records their total multiplicity, not their separation.

3.2 Chinese remainders, Hermite jets, and idempotents

Let P be monic as in (3.1), and put $A_i = (z - \alpha_i)^{m_i}$, $P_i = P/A_i$. The inverse of P_i modulo A_i is the finite Taylor polynomial at α_i of the rational function $1/P_i$,

$$C_i(z) = \sum_{j=0}^{m_i-1} \frac{1}{j!} \left(\frac{d^j}{dz^j} \frac{1}{P_i(z)} \right)_{z=\alpha_i} (z - \alpha_i)^j, \quad (3.4)$$

whose denominators are nonzero because distinct α_i are separated as field elements, even when their differences are infinitesimal.

Theorem 3.2 (Finite Hermite interpolation and CRT). *The map*

$$K[z]/(P) \longrightarrow \prod_{i=1}^s K[u]/(u^{m_i}), \quad [H] \longmapsto \left[\sum_{j < m_i} \frac{H^{(j)}(\alpha_i)}{j!} u^j \right]_i \quad (3.5)$$

is an isomorphism: arbitrary jets of orders less than m_i at the α_i are realized by a unique polynomial interpolant of degree less than n . The classes $E_i = [P_i C_i] \in K[z]/(P)$ are orthogonal idempotents with sum 1 and give the identities of the factors. When all roots are simple this is the Lagrange formula

$$H(z) = \sum_{i=1}^n H(\alpha_i) \frac{P(z)}{(z - \alpha_i) P'(\alpha_i)}, \quad \deg H < n. \quad (3.6)$$

Proof. The finite Taylor identity gives $P_i C_i \equiv 1 \pmod{A_i}$ and $P_i C_i \equiv 0 \pmod{A_j}$ for $j \neq i$, so the remainder modulo P of $\sum_i P_i C_i J_i(z - \alpha_i)$ realizes prescribed jets J_i . A polynomial in the kernel is divisible by each pairwise coprime A_i , hence by their product P ; unique division by P supplies the degree bound and uniqueness. The idempotent identities hold in each factor and therefore in the quotient. $\square$

This is a genuinely finite interpolation theorem: only finitely many coefficient supports are combined, so it meets none of the infinite symmetric-support obstruction of [6]. Its inverse can nevertheless introduce inverse powers of small separations — already interpolation at 0 and t^ρ uses z/t^ρ to realize the values 0, 1 — and the denominators in (3.6) may be infinitely large after inversion. No separation lower bound in ordinary real units is required.

3.3 Vieta, Newton sums, resultants and discriminants

Write $P(z) = z^n + c_1 z^{n-1} + \cdots + c_n$ and $s_k = \sum_j \alpha_j^k$. Expanding the finite product gives $c_k = (-1)^k e_k(\alpha_1, \dots, \alpha_n)$, and the Newton identities

$$s_k + c_1 s_{k-1} + \cdots + c_{k-1} s_1 + k c_k = 0 \quad (1 \leq k \leq n), \quad (3.7)$$

$$s_k + c_1 s_{k-1} + \cdots + c_n s_{k-n} = 0 \quad (k > n) \quad (3.8)$$

follow by expanding (3.2) at infinity as a formal Laurent series in z^{-1} and comparing finitely many coefficients after multiplying by P . No infinite analytic evaluation is involved.

For $P = a_n \prod_i (z - \alpha_i)$ and $Q = b_m \prod_j (z - \beta_j)$ define

$$\text{Res}(P, Q) = a_n^m \prod_{i=1}^n Q(\alpha_i) = a_n^m b_m^n \prod_{i,j} (\alpha_i - \beta_j), \quad (3.9)$$

symmetric up to $(-1)^{nm}$, a polynomial in the coefficients, and equal to the Sylvester determinant. For monic P we will always use the multiplication-determinant definition $\text{Res}(P, Q) = \det(m_Q : K[z]/(P) \rightarrow K[z]/(P))$, justified in [theorem 14.2](#). Set

$$\text{Disc}(P) = a_n^{2n-2} \prod_{i < j} (\alpha_i - \alpha_j)^2, \quad \text{Disc}(P) = (-1)^{n(n-1)/2} \text{Res}(P, P') \quad \text{for monic } P, \quad (3.10)$$

which vanishes exactly at a repeated root. When the relevant products do not vanish,

$$v(\text{Res}(P, Q)) = m v(a_n) + n v(b_m) + \sum_{i,j} v(\alpha_i - \beta_j), \quad (3.11)$$

$$v(\text{Disc}(P)) = (2n-2)v(a_n) + 2 \sum_{i < j} v(\alpha_i - \alpha_j) = \sum_i d_i \quad (P \text{ monic, squarefree}). \quad (3.12)$$

These quantities record collisions across all scales, not merely ordinary coincidence of standard parts. The resultant vanishes exactly when P, Q have a common root, equivalently when the finite Bézout map is singular; this is a coefficient-level test and not a promise of an effective algorithm for equality of arbitrary infinitely supported surreal coefficients.

4 Ordered (modulus) root geometry

4.1 Root bounds at arbitrary surreal radii

Proposition 4.1 (Cauchy bounds and a radial bound). *Let $P(z) = a_n z^n + \cdots + a_0$ with $a_n \neq 0$ and $n \geq 1$. Every root satisfies*

$$|\alpha| < 1 + \max_{j < n} |a_j/a_n|. \quad (4.1)$$

If $a_0 \neq 0$ it also satisfies $|\alpha| > (1 + \max_{1 \leq j \leq n} |a_j/a_0|)^{-1}$. With $M = \max_{j < n} |a_j/a_n|^{1/(n-j)}$ one has $|\alpha| \leq 2M$, and if $M = 0$ all roots are zero.

Proof. Divide by a_n and set $A = \max_{j < n} |a_j/a_n|$; the case $A = 0$ is immediate. For $r = |\alpha| \geq 1 + A > 1$ the equation $P(\alpha) = 0$ and the finite geometric identity give $r^n \leq A(r^n - 1)/(r - 1) \leq r^n - 1$, a contradiction. Apply this to the reciprocal polynomial $z^n P(1/z)/a_0$ for the lower bound. Finally if $r > 2M > 0$ then $\sum_{j < n} |a_j/a_n| r^{j-n} \leq \sum_{k=1}^n 2^{-k} < 1$, so the leading term cannot be cancelled; the positive roots defining M exist in the real closed field. $\square$

The bounds hold literally for infinite or infinitesimal coefficients, and no Archimedean estimate is used — the geometric sum is finite, and the argument is valid for an infinite surreal r as well as a finite one. Applying them to $P(a + rz)$ gives corresponding bounds at any center and scale.

4.2 Gauss–Lucas with surreal convex weights

For a finite $S \subseteq K$ the F -convex hull is

$$\text{conv}_F(S) = \left\{ \sum_{r \in S} \theta_r r : \theta_r \in F, \theta_r \geq 0, \sum_r \theta_r = 1 \right\}.$$

Allowing surreal, rather than only ordinary real, weights is essential: restricting them to ordinary real numbers would be a different and generally too small a hull. This is a finite-dimensional algebraic notion of convexity; no topological compactness of a proper-class set is involved. Every order disk and every order half-plane is convex, by the triangle inequality and linearity of the real part.

Theorem 4.2 (Surcomplex Gauss–Lucas). *Every root of P' lies in the F -convex hull of the roots of a nonconstant $P \in \mathbf{SC}[z]$. Explicitly, if w is a critical point with $P(w) \neq 0$ then*

$$w = \frac{\sum_i \frac{m_i \alpha_i}{|w - \alpha_i|^2}}{\sum_i \frac{m_i}{|w - \alpha_i|^2}}, \quad (4.2)$$

all weights being strictly positive elements of F . The same convex hull contains the zeros of every nonzero higher derivative.

Proof. At a nonroot critical point the logarithmic derivative (3.2) vanishes. Since $1/(w - \alpha_i) = (\bar{w} - \bar{\alpha}_i)/|w - \alpha_i|^2$, conjugating gives $\sum_i m_i(w - \alpha_i)/|w - \alpha_i|^2 = 0$; every weight is positive, their sum is nonzero and may be divided out, giving (4.2). A critical point that is itself a root is already in the hull. Repeating for successive derivatives and using convexity gives the last assertion. $\square$

The displayed proof is finite and algebraic; it does not call on compactness of a convex hull or a topological separation theorem. Its classical analogue is Gauss–Lucas, and modern formal developments of the ordinary theorem and related root geometry are available [15]. Neither conclusion describes how critical points are distributed among infinitesimal clusters; that more precise question is answered in section 11.

Proposition 4.3 (Weighted incomplete polynomials, and a converse). *For $\alpha_1, \dots, \alpha_n \in K$ and weights $\lambda_j \geq 0$ in F with $\sum_j \lambda_j > 0$, every root of $Q_\lambda(z) = \sum_{j=1}^n \lambda_j \prod_{k \neq j} (z - \alpha_k)$ belongs to $\text{conv}_F\{\alpha_1, \dots, \alpha_n\}$. Conversely, for $n \geq 2$ the union of these root sets as λ varies is exactly that convex hull.*

Proof. At a root distinct from all α_j , divide by $\prod_j (z - \alpha_j)$ and repeat the proof of theorem 4.2 with weights $\lambda_j/|z - \alpha_j|^2$. For the converse let $w = \sum_j \theta_j \alpha_j$ with $\theta_j \geq 0$, $\sum_j \theta_j = 1$. If w is not a listed root, take $\lambda_j = \theta_j |w - \alpha_j|^2$; then $\sum_j \lambda_j/(w - \alpha_j) = \sum_j \theta_j (\bar{w} - \bar{\alpha}_j) = 0$. At a listed root α_i , choose a nonzero weight at an index $j \neq i$, so that the incomplete product contains the factor $z - \alpha_i$. $\square$

This is an ordered-field form of a classical extension of Gauss–Lucas [18]; the proof exhibits precisely why it survives for surreal coefficients. It will be used again in theorem 11.6, where the branch polynomial at a node of the root-cluster tree is a weighted incomplete polynomial of exactly this kind.

4.3 Jensen disks for real-surreal coefficients

Theorem 4.4 (Surcomplex Jensen disk theorem). *Suppose $P \in F[z]$ is nonconstant. Every nonreal critical point belongs to at least one closed order disk having a pair of conjugate nonreal roots as endpoints of a diameter; equivalently, for some root pair $a \pm ib$ with $b \neq 0$ the critical point w satisfies $|w - a| \leq |b|$.*

Proof. Conjugation preserves the root multiset. A critical point that is a root lies on the appropriate boundary, so assume $w = x + iy$ is not a root and $y \neq 0$. For a real root r , $y^{-1} \operatorname{Im}(w - r)^{-1} = -|w - r|^{-2} < 0$. For a conjugate pair $a \pm ib$,

$$\frac{1}{y} \operatorname{Im} \left(\frac{1}{w - a - ib} + \frac{1}{w - a + ib} \right) = \frac{-2((x - a)^2 + y^2 - b^2)}{|w - a - ib|^2 |w - a + ib|^2}. \quad (4.3)$$

If w were outside every indicated closed disk, every contribution to $y^{-1} \operatorname{Im}(P'/P)(w)$, with its positive multiplicity, would be negative, and their finite sum could not vanish. This contradicts $P'(w) = 0$. $\square$

The nonconstant hypothesis is necessary: for $P = 1$ the derivative is zero, so every nonreal point is a zero of P' while there is no root pair to define a disk. In particular a nonconstant real-rooted polynomial has only real critical points. The assertion concerns F -valued disks; it is not a statement about ordinary pictures obtained by discarding infinitesimals. Neither of the other two merged manuscripts proves a Jensen disk theorem; one of them proves [theorem 6.4](#) in its place.

5 A sharp second-moment inequality for critical points

The ordinary Schoenberg inequality is established mathematics, proved with matrix differentiators and related spectral methods [16, 17]. We give an algebraic proof over a general real closed field, including the equality case, so that no Hilbert-space completeness assumption is hidden in a transfer. This section is the contribution of [2] alone; neither of the other two merged manuscripts contains any second-moment or matrix-compression material.

5.1 A matrix whose eigenvalues are the critical points

Let P be monic of degree $n \geq 2$ with roots $\alpha_1, \dots, \alpha_n$ repeated by multiplicity, and set $D = \operatorname{diag}(\alpha_1, \dots, \alpha_n)$, $e = n^{-1/2}(1, \dots, 1)^T$, $\Pi = I - ee^*$, where $*$ is conjugate transpose. Algebraic Gram–Schmidt over F gives a real $n \times (n - 1)$ matrix U with $U^*U = I$, $UU^* = \Pi$. Put $B = U^*DU$.

Lemma 5.1 (Differentiating compression). *The characteristic polynomial of B is $P'(z)/n$.*

Proof. The unitary $(e \ U)$ turns $zI - D$ into a block matrix whose lower-right block is $zI_{n-1} - B$, so its upper-left cofactor is $\det(zI_{n-1} - B)$. In the original coordinates that cofactor is $e^* \operatorname{adj}(zI - D)e$, which for diagonal D equals $n^{-1} \sum_i \prod_{j \neq i} (z - \alpha_j) = P'(z)/n$. All expressions are polynomial identities, so no invertibility at a root of P is assumed. $\square$

Lemma 5.2 (Finite-dimensional Schur inequality). *If $A \in \operatorname{Mat}_m(K)$ has eigenvalues $\lambda_1, \dots, \lambda_m$ with multiplicity, then $\sum_j |\lambda_j|^2 \leq \operatorname{Tr}(AA^*)$, with equality exactly when A is normal.*

Proof. Algebraic closedness provides an eigenvector; normalize it by the positive square root of its squared norm, extend to an orthonormal basis by Gram–Schmidt and induct, giving a unitary upper triangularization. In that basis $\operatorname{Tr}(AA^*)$ is the sum of squared moduli of all entries, while the left side is the sum over the diagonal; the difference is the sum over the strict upper triangle and vanishes exactly when those entries vanish. A diagonal matrix is normal and normality is unitarily invariant; conversely an upper-triangular normal matrix is diagonal, since equality of the $(1, 1)$ entries of TT^* and T^*T kills the first row above the diagonal and induction handles the rest. $\square$

5.2 The inequality and its equality case

Theorem 5.3 (Sharp surcomplex Schoenberg inequality). *Let $w_1, \dots, w_{n-1}$ be the roots of P' with multiplicity. Then*

$$\sum_{j=1}^{n-1} |w_j|^2 \leq \frac{n-2}{n} \sum_{i=1}^n |\alpha_i|^2 + \frac{1}{n^2} \left| \sum_{i=1}^n \alpha_i \right|^2, \quad (5.1)$$

equivalently, with the root centroid $c = n^{-1} \sum_i \alpha_i$,

$$\boxed{\sum_{j=1}^{n-1} |w_j - c|^2 \leq \frac{n-2}{n} \sum_{i=1}^n |\alpha_i - c|^2.} \quad (5.2)$$

Equality holds if and only if all the roots lie on an affine F -line in K ; the one-point configuration counts as collinear.

Proof. Apply lemma 5.2 to B and use lemma 5.1. Cyclic invariance of the finite trace gives

$$\mathrm{Tr}(BB^*) = \mathrm{Tr}(D\Pi D^*\Pi) = \left(1 - \frac{2}{n}\right) \sum_i |\alpha_i|^2 + \frac{1}{n^2} \left| \sum_i \alpha_i \right|^2,$$

which is (5.1); Vieta for P' gives $\sum_j w_j = (n-1)c$, and replacing every root by its difference from c gives (5.2).

For equality put $v = \Pi D e$ and $u = \Pi D^* e = \bar{v}$, the last because Π is real. On $e^\perp$, expanding $\Pi = I - ee^*$ and using $DD^* = D^*D$ represents $BB^* - B^*B$ by $uu^* - vv^*$. If $v = 0$ all roots equal c . Otherwise equality of these two rank-one matrices forces $u = \zeta v$ with $\zeta \bar{\zeta} = 1$, their common nonzero image being one-dimensional. The coordinates of v are $(\alpha_i - c)/\sqrt{n}$, so normality is equivalent to $\overline{\alpha_i - c} = \zeta(\alpha_i - c)$ for all i . Choosing a nonzero $d = \alpha_j - c$ and dividing shows every ratio $(\alpha_i - c)/d$ is fixed by conjugation, hence lies in F , so all roots lie on $c + dF$. Conversely that collinearity makes $\bar{v}$ a unit-modulus multiple of v , hence B normal. $\square$

The coefficient $(n-2)/n$ cannot be improved, since nonconstant collinear configurations attain equality. Both sides may be infinite, finite or infinitesimal surreal quantities; they are compared in the ordered field, not after taking standard parts. The roots $1, i, -1, -i$ of $z^4 - 1$ have all critical points at zero, so the inequality is strict; multiplying every root by a nonzero surcomplex scale gives an equally valid strict example at that scale. The matrix proof uses the positive definite form on the finite-dimensional space K^n and algebraic square roots; it does not assert completeness of an infinite-dimensional surcomplex Hilbert space.

6 Real-closed-field transfer: Rouché on order circles, and Hermite signatures

6.1 Exactly what transfer permits

Tarski's theorem gives quantifier elimination for real closed ordered fields, so a first-order sentence in the ordered-field language true in $\mathbb{R}$ is true in every real closed field [11, 12]. For any *fixed* degree bound a complex-coefficient polynomial is finitely many pairs of real coordinates; polynomial identities, circle equations and squared-modulus inequalities are then formulas in that language. Root counts with multiplicity are also encodable: introduce a finite list of roots, impose the coefficient

identities of a complete factorization, and impose by a finite disjunction that exactly k entries of the list lie in the specified disk. This counts repeated entries rather than only distinct points; degrees below a chosen bound are handled by a finite disjunction on the last nonzero coefficient.

This is a theorem *scheme*, one sentence for each fixed finite degree bound. It does not transfer unrestricted quantification over functions, sequences, arbitrary subsets, the standard natural numbers as a distinguished definable set, or all fine-continuous functions. It also does not make a proper class into a first-order set-sized model: transfer is applied to F_Γ , after all parameters have been placed in that field. Constructive real-algebraic winding-number and Sturm-chain methods give another route to root counting over real closed fields [14]; we do not need that machinery for the implications stated here.

Theorem 6.1 (Rouché on an order-modulus circle). *Let $P, Q \in \mathbf{SC}[z]$ be nonzero, $a \in \mathbf{SC}$, $r \in \mathbf{No}_{>0}$. Suppose*

$$|Q(z) - P(z)| < |P(z)| \quad \text{for every } z \in \mathbf{SC} \text{ with } |z - a| = r. \quad (6.1)$$

Then P and Q have the same number of roots in $D(a, r)$, counted with multiplicity, and neither has a root on the circle.

Proof. Fix the two finite degrees. Hypothesis (6.1) is a first-order formula: quantify the two real coordinates of z , write the circle as $(\Re z - \Re a)^2 + (\Im z - \Im a)^2 = r^2$, and square both nonnegative sides. The conclusion is first order at those fixed degrees by the encoding above. The resulting universally quantified implication is true over $\mathbb{R}$ by ordinary polynomial Rouché, hence in every real closed field. Choose F_Γ containing the real and imaginary parts of the coefficients, a and r ; the hypothesis holds there because it holds in $\mathbf{SC}$, and by proposition 2.1 the root lists already contain every surcomplex root, so the conclusion is the desired count in $\mathbf{SC}$. Finally the strict inequality excludes a boundary root of P , and a boundary root of Q would make the two sides equal. $\square$

This is a proof by transfer from a classical theorem, and is identified as such. It does not define an integral along a fine-continuous real-parameter surreal circle. It is also not the same statement as the leading-coefficient Rouché theorem on ordinary thickened domains in [4]: its hypothesis concerns the *full* order circle at the chosen surreal scale.

Corollary 6.2 (Monomial dominance). *Write $P(a + Z) = \sum_{j=0}^n p_j Z^j$. If for some k and $r > 0$*

$$|p_k| r^k > \sum_{j \neq k} |p_j| r^j, \quad (6.2)$$

then P has exactly k roots in $D(a, r)$ counted with multiplicity, and none on the boundary. The case $k = 0$ (no roots) is included.

Proof. On $|z - a| = r$ the difference between $P(z)$ and $p_k(z - a)^k$ has modulus at most the right side of (6.2). Apply theorem 6.1 with that monomial as reference; it has exactly k roots in the disk, counting multiplicity at a . $\square$

No strict inequality among the valuations of the terms is required. The criterion can distinguish two terms at the same valuation by their leading ordinary moduli — a thing the valuation criterion of corollary 8.3 cannot do, and the clearest illustration of why the two are not interchangeable.

Example 6.3 (Testing only ordinary circle points is insufficient). Let $t = \omega^{-1}$ and put

$$b = \frac{1+it}{\sqrt{1+t^2}}, \quad \alpha = (1-t^3)b, \quad \beta = (1+t^3)b, \quad P(z) = z - \alpha, \quad Q(z) = z - \beta.$$

Then $|b| = 1$, $|\alpha| = 1 - t^3 < 1$ and $|\beta| = 1 + t^3 > 1$, so P and Q have different root counts in the unit order disk. Nevertheless $|Q(\zeta) - P(\zeta)| < |P(\zeta)|$ at every *ordinary* complex unit point ζ : for $\zeta \neq 1$ one has $v(P(\zeta)) = 0$, while at $\zeta = 1$ it is 1, the leading displacement of b being it ; the error has valuation 3 in both cases. At the nonordinary unit point b , however, $|P(b)| = t^3$ and $|Q(b) - P(b)| = 2t^3$, so the hypothesis genuinely fails there. The exact quantifier in (6.1) cannot be replaced by sampling ordinary points.

6.2 Hermite trace forms count real roots

Let R be a real closed field, $P \in R[z]$ monic of degree n , $A = R[z]/(P)$, and for $q \in A$

$$T_q(f, g) = \text{Tr}_A(m_{qfg}). \quad (6.3)$$

A symmetric matrix over a real closed field is diagonalizable by congruence; its *signature* is the number of positive minus the number of negative diagonal entries, zeros ignored, well defined by Sylvester's law of inertia. In the monomial basis its entries are $\text{Tr}(q(C_P)C_P^{i+j})$, where C_P is the companion matrix.

Theorem 6.4 (Hermite's signature criterion over surreal real closed fields). *With the above notation,*

$$\text{sig } T_q = \sum_{\substack{r \in R \\ P(r)=0}} \text{sgn } q(r), \quad (6.4)$$

the sum over distinct real roots. In particular $\text{sig } T_1$ is their number, the rank of T_1 is the number of distinct roots over $R[i]$, and T_1 is positive semidefinite if and only if every root is real. For $a < b$ with $P(a)P(b) \neq 0$ and $q(z) = (z - a)(b - z)$,

$$\#\{r \in (a, b) : P(r) = 0\} = \frac{1}{2}(\text{sig } T_1 + \text{sig } T_q), \quad (6.5)$$

again counting distinct roots; apply it to the squarefree multiplicity factors for a count with multiplicity.

Proof. Factor P over R into powers of distinct real linear factors and irreducible quadratics; Chinese remaindering makes the summands orthogonal. On a local factor at a real root r of multiplicity m , multiplication by h has trace $mh(r)$, since its nilpotent part is triangular with zero diagonal; the form therefore factors through the one-dimensional residue field as $mq(r)uv$, with signature $\text{sgn } q(r)$. For a nonreal conjugate pair the residue field is $R[i]$ and, with $q(r) = a_0 + ib_0$, the residue form is m times $\begin{pmatrix} 2a_0 & -2b_0 \\ -2b_0 & -2a_0 \end{pmatrix}$, of determinant $-4(a_0^2 + b_0^2) < 0$ when $q(r) \neq 0$: signature zero, rank two.

If $q(r) = 0$ it is zero. Nilpotent directions lie in the kernel in both cases. Adding signatures gives (6.4), and $q = 1$ gives the remaining assertions; (6.5) follows since $q > 0$ exactly on (a, b) . $\square$

This classical signature method is related to the residue and Bézoutian forms discussed in [23]; it is *not* the nondegenerate residue pairing of section 14. The two forms are compared explicitly in theorem 14.6: the trace pairing degenerates exactly at a repeated root, the residue pairing never does. Only one of the three merged manuscripts proves this criterion; the one that proves theorem 4.4 proves no signature criterion, and neither subsumes the other.

7 Exact root loci under coefficient uncertainty

The next theorem gives a *sharp* description of the reachable root set under bounded coefficient errors, not merely a sufficient bound, and it exposes the structural contrast between the two geometries. Weighted coefficient-uncertainty root regions are called *pseudozero sets* in ordinary polynomial analysis, and root neighbourhoods of this type have established applications [22]; the argument below is given in full for the surcomplex settings used here. This section is the contribution of [1] alone; neither of the other two merged manuscripts has an uncertainty-locus section.

Fix a monic degree- n polynomial $P \in \mathbf{SC}[z]$ and a set of perturbable indices $J \subseteq \{0, \dots, n-1\}$, and consider $Q(z) = P(z) + \sum_{j \in J} e_j z^j$, so that all perturbations retain the degree and leading coefficient.

Theorem 7.1 (Exact uncertainty root sets in two geometries). *Fix $z \in \mathbf{SC}$.*

Order-modulus version. *Given $\eta_j \in \mathbf{No}_{\geq 0}$, such a Q with $Q(z) = 0$ and $|e_j| \leq \eta_j$ for every $j \in J$ exists if and only if*

$$|P(z)| \leq \sum_{j \in J} \eta_j |z^j|. \quad (7.1)$$

Valuation version. *Given $\nu_j \in \Gamma$, such a Q with $Q(z) = 0$ and $v(e_j) \geq \nu_j$ for every $j \in J$ exists if and only if*

$$v(P(z)) \geq \min_{j \in J} (\nu_j + v(z^j)). \quad (7.2)$$

Here $z^0 = 1$, $v(0) = +\infty$, and the minimum of an empty set is $+\infty$. If the minimum on the right is finite, replacing every bound by $v(e_j) > \nu_j$ replaces $\geq$ by $>$ in (7.2).

Proof. The triangle inequality gives necessity of (7.1). Let S be its right side. If $S = 0$ the inequality forces $P(z) = 0$ and all errors may be taken zero. If $S > 0$ put, for $z^j \neq 0$,

$$e_j = -\frac{P(z)}{S} \eta_j \frac{\overline{z^j}}{|z^j|}, \quad (7.3)$$

and $e_j = 0$ otherwise. Then $|e_j| \leq \eta_j$ and $\sum_{j \in J} e_j z^j = -\frac{P(z)}{S} \sum_{j \in J} \eta_j |z^j| = -P(z)$, which proves sufficiency, including $z = 0$ with the stated convention for z^0 . The phases in (7.3) are algebraic unit-modulus factors; no global argument or trigonometric branch is needed.

For the valuation version, the valuation inequality for a finite sum gives necessity. If the minimum is finite, choose j_0 attaining it and take $e_{j_0} = -P(z)/z^{j_0}$ and $e_j = 0$ otherwise; the assumed bound says exactly that this candidate coefficient has the required valuation. If the minimum is $+\infty$, every perturbation monomial vanishes at z or there are none, so $Q(z) = P(z)$ and the condition correctly reduces to $P(z) = 0$. For a finite minimum the same reasoning with strict inequalities gives the last assertion, discarding monomials vanishing at z when checking necessity: the minimum of finitely many quantities each strictly above its own bound is strictly above the minimum of those bounds. $\square$

The contrast is exact, not an analogy based on a heuristic ultrametric norm. The order formula uses a *sum* of available magnitudes, so aligned phases let several coefficients cooperate, and the cooperation is realized explicitly by (7.3). The valuation formula uses a *minimum*, and one suitably chosen coefficient always suffices. The proof permits infinite weights and arbitrary surreal centers because every operation is finite.

Example 7.2 (Constant-term perturbations). For $P(z) = z^m - t$ with $m \geq 1$, a permitted constant error of modulus at most η gives exactly the region $|z^m - t| \leq \eta$, and a constant error of valuation at least ν gives exactly $v(z^m - t) \geq \nu$. When $\nu > 1$ the root residues at scale $1/m$ cannot change, although their higher terms can. At $\nu = 1$ the choice $e_0 = t$ changes the polynomial to z^m and makes all roots collide at zero. The comparison anticipates the strictness in [example 13.5](#).

8 The initial polynomial at a center and a scale

8.1 Weighted Gauss valuation

Fix a nonzero $P \in K[z]$, a center $a \in K$ and a scale exponent $\rho \in \Gamma$. Its finite Taylor expansion is $P(a + Z) = \sum_{j=0}^n b_j Z^j$ with $b_j = P^{(j)}(a)/j!$, and we define

$$w_{a,\rho}(P) = \min_{b_j \neq 0} \{v(b_j) + j\rho\} =: \lambda, \quad I_{a,\rho}(P)(X) = \text{st}(t^{-\lambda} P(a + t^\rho X)) \in \mathbb{C}[X], \quad (8.1)$$

the standard part taken coefficientwise, with $w_{a,\rho}(0) = +\infty$. All coefficients inside the parentheses have nonnegative valuation and at least one is a unit, so the *initial polynomial* $I_{a,\rho}(P)$ is nonzero. It need not be monic and may have degree smaller than n : terms of strictly larger weighted valuation disappear under reduction. This definition uses no limiting process and is meaningful for every $\rho \in \Gamma$, including negative thresholds and non-Archimedean surreal exponents. Put

$$J_{a,\rho}(P) = \{j : v(b_j) + j\rho = \lambda\}, \quad j_- = \min J_{a,\rho}(P), \quad j_+ = \max J_{a,\rho}(P), \quad (8.2)$$

so that $\text{ord}_0 I_{a,\rho}(P) = j_-$ and $\deg I_{a,\rho}(P) = j_+$.

Lemma 8.1 (Multiplicativity). *For nonzero $P, Q \in K[z]$,*

$$w_{a,\rho}(PQ) = w_{a,\rho}(P) + w_{a,\rho}(Q), \quad I_{a,\rho}(PQ) = I_{a,\rho}(P) I_{a,\rho}(Q),$$

and $w_{a,\rho}(P + Q) \geq \min\{w_{a,\rho}(P), w_{a,\rho}(Q)\}$.

Proof. Normalize both as in (8.1). Their reductions are nonzero and their product is nonzero in the domain $\mathbb{C}[X]$, so the normalized product has least coefficient valuation exactly zero; this proves both multiplicative identities. The additive inequality holds coefficient by coefficient before reduction. $\square$

No cancellation assumption about the original coefficients is needed: cancellation is already accounted for by computing actual valuations and then multiplying nonzero reductions. This weighted valuation is the Gauss valuation; unlike the modulus it is ultrametric, and no supremum over a topological circle is needed to define it.

8.2 The exact root formula

Theorem 8.2 (Initial polynomial and all residue-direction root counts). *Write $P = a_n \prod_{i=1}^n (z - \alpha_i)$ with multiplicities, over K . For every $a \in K$ and $\rho \in \Gamma$ there is a nonzero constant $C_{a,\rho} \in \mathbb{C}$ with*

$$I_{a,\rho}(P)(X) = C_{a,\rho} \prod_{v(\alpha_i - a) \geq \rho} \left(X - \text{st} \frac{\alpha_i - a}{t^\rho} \right), \quad (8.3)$$

and moreover

$$w_{a,\rho}(P) = v(a_n) + \sum_{i=1}^n \min\{\rho, v(\alpha_i - a)\}, \quad (8.4)$$

with the convention $\min(\rho, +\infty) = \rho$ for roots at a . Consequently, counting multiplicity,

$$\#\{\alpha_i \in B_{\geq}(a, \rho)\} = \deg I_{a,\rho}(P) = j_+, \quad (8.5)$$

$$\#\{\alpha_i \in B_{>}(a, \rho)\} = \text{ord}_0 I_{a,\rho}(P) = j_-, \quad (8.6)$$

$$\#\{i : v(\alpha_i - a) = \rho\} = j_+ - j_-. \quad (8.7)$$

For each $c \in \mathbb{C}$, the number of roots of P in the residue-direction ball $a + t^\rho(c + \mathfrak{m}_K)$ equals the multiplicity of c as a root of $I_{a,\rho}(P)$.

Proof. By lemma 8.1 both the weighted valuation and the initial polynomial are multiplicative, so it suffices to treat one linear factor. If $v(\alpha - a) \geq \rho$, write $\beta = (\alpha - a)/t^\rho \in \mathcal{O}_K$; then $a + t^\rho X - \alpha = t^\rho(X - \beta)$, of weighted valuation ρ and normalized reduction $X - \text{st}(\beta)$, which is X exactly when $v(\alpha - a) > \rho$. If $v(\alpha - a) < \rho$, put $\delta = v(\alpha - a)$ and $\ell = \text{lc}(\alpha - a)$; normalizing $t^\rho X - (\alpha - a)$ by $t^{-\delta}$ gives reduction the nonzero constant $-\ell$ and weighted valuation δ . The leading scalar a_n contributes its leading complex coefficient and its valuation. Multiplying these finitely many results proves (8.3) and (8.4). A finite normalized root reduces to 0 exactly when its valuation is positive and to c exactly when it lies in $c + \mathfrak{m}_K$; degree, order at zero and multiplicity at c then give all the counts, and the nonzero coefficients of $I_{a,\rho}(P)$ occur exactly at the minimizing indices, which identifies $j_{\pm}$. $\square$

The theorem packages all roots of the chosen ball into a single ordinary complex polynomial. Roots outside that ball contribute only a nonzero scalar to the reduction, but their valuations still contribute to (8.4). It simultaneously allows a to be infinite, t^ρ to be infinite or infinitesimal, and Γ to have arbitrary ordered-group rank. It is a finite valued-field argument, not an appeal to analytic compactness of a ball, and a single reduction carries more information than an unlabelled Newton polygon, since it also distinguishes directions *within* a scale.

Corollary 8.3 (Valuation Rouché: preservation of the whole initial polynomial). *Let $P \neq 0$ and $E \in K[z]$ satisfy*

$$w_{a,\rho}(E) > w_{a,\rho}(P), \quad w_{a,\rho}(0) = +\infty. \quad (8.8)$$

Then $w_{a,\rho}(P + E) = w_{a,\rho}(P)$ and $I_{a,\rho}(P + E) = I_{a,\rho}(P)$; consequently P and $P + E$ have equal root counts in $B_{\geq}(a, \rho)$, in $B_{>}(a, \rho)$, and in every residue-direction ball at that center and scale.

Proof. Normalize by $t^{-w_{a,\rho}(P)}$ after substituting $a + t^\rho X$; every coefficient of the scaled E then has positive valuation, so its reduction is zero and cannot cancel the nonzero reduction of P . Apply theorem 8.2. $\square$

The degrees of P and $P + E$ need not agree, and additional roots of the higher-degree polynomial may lie outside the selected ball — this is exactly the situation of example 2.8; only the specified ball counts are asserted. The strict inequality is necessary for the general statement: $P = z$ and $P + E = z + 1$ at scale zero have different residue directions although the error has the same, rather than a smaller, weighted size. The coefficient test (8.8) has the benefit of not quantifying over boundary points at all. The modulus and valuation versions of Rouché have different hypotheses and different balls, and neither is identified with the contour theorem on an ordinary thickening in [4]; this proof uses no transfer and no version of contour integration.

9 Newton profiles over an arbitrary ordered exponent group

9.1 The polygon without an Archimedean assumption

At center zero put $\phi_P(\rho) = \min_j \{v(a_j) + j\rho\}$, a minimum of finitely many affine expressions in the ordered divisible group Γ . One organizes it by its finite breakpoints without embedding Γ in $\mathbb{R}$: candidate breakpoints are $(v(a_j) - v(a_k))/(k - j)$, which lie in Γ by divisibility, and only those at which at least two indices attain the minimum matter. Equivalently, considering the finitely many coefficient points $(j, v(a_j))$ with $a_j \neq 0$, a *lower face of slope $-\rho$* is specified *algebraically* by the set of indices minimizing $v(a_j) + j\rho$. This definition works even when Γ cannot be embedded in $\mathbb{R}$; we do not require a literal Euclidean drawing of its vertical axis.

Theorem 9.1 (Newton’s root-valuation rule). *Over K , at a threshold $\rho \in \Gamma$ (rather than the added value $+\infty$), the number of nonzero roots of valuation ρ is the difference between the largest and smallest indices attaining $\phi_P(\rho)$, and every valuation of a nonzero root is such a breakpoint. Equivalently, a lower face joining indices $j < k$ with slope $-\rho$ accounts for $k - j$ roots of valuation ρ , and the residue polynomial of that face determines their leading complex coefficients with multiplicity. Roots equal to the center contribute their exact multiplicity separately, as roots of valuation $+\infty$.*

Proof. The count at a fixed scale is (8.7). If a nonzero root has valuation ρ , that count is positive, so at least two indices attain the minimum; conversely a positive index difference produces exactly that many roots by the same formula. A lower face is precisely a set of coefficient points on the line $y + \rho x = \phi_P(\rho)$ with all others above it, so its horizontal span is the difference of the extreme active indices. The leading complex coefficients are read from (8.3). Alternatively, (8.4) expresses $\phi_P(\rho) = v(a_n) + \sum_i \min\{\rho, v(\alpha_i)\}$, whose slope changes at precisely the root valuations, with the same multiplicities. Both arguments use only order and rational scalar multiplication in Γ . $\square$

The familiar Newton polygon is therefore a finite encoding of the root-valuation multiset. The standard rank-one analytic version and valuation polygons are discussed in [19, §3] and in [20]; the proof above requires neither a real-valued norm nor completeness. The one-variable tropical condition that the minimum is attained at least twice is thus sufficient as well as necessary for a nonzero root at that valuation — a consequence of algebraic closedness together with the exact count, *not* an additional convergence assertion about a Newton algorithm.

Example 9.2 (A cubic with two root scales). Let $t = \omega^{-1}$ and $P(z) = z^3 - t^2z - t^5$. The nonzero coefficient points are $(0, 5), (1, 2), (3, 0)$, so $\phi_P(\rho) = \min\{5, 2 + \rho, 3\rho\}$: at $\rho = 1$ the active indices are 1, 3, giving two roots of valuation one, and at $\rho = 3$ they are 0, 1, giving one root of valuation three. Explicitly

$$\begin{aligned} t^{-3}P(tX) &= X^3 - X - t^2, & I_{0,1}(P) &= X^3 - X = X(X - 1)(X + 1), \\ t^{-5}P(t^3X) &= t^4X^3 - X - 1, & I_{0,3}(P) &= -X - 1. \end{aligned}$$

The three residue directions at scale one are 0, 1, -1 ; the zero direction has higher valuation and is resolved at the next relevant scale. The simple-root recursion of corollary 12.2 gives

$$\alpha_+ = t + \frac{1}{2}t^3 - \frac{3}{8}t^5 + O(t^7), \quad \alpha_- = -t + \frac{1}{2}t^3 + \frac{3}{8}t^5 + O(t^7), \quad \alpha_0 = -t^3 - t^7 + O(t^{11}).$$

Here $O(t^k)$ means an integer-power formal Hahn series supported in exponents at least k , not a fine-topological limiting estimate and not a numerical asymptotic assertion obtained by varying t . The critical points are exactly $\pm t/\sqrt{3}$, since $P' = 3z^2 - t^2$; they lie in none of the three root submonads at scale one, although both lie in their common valuation ball.

Example 9.3 (A scale beyond all integer multiples). In a divisible workspace containing 1 and ω , take $P(z) = (z - t)(z - t^\omega)$. Its two roots have valuations 1 and ω ; the Newton profile has both breakpoints and [theorem 9.1](#) applies unchanged. There is no integer N with $N > \omega$, and an iteration that only raises a valuation through the sequence 1, 2, 3, ... cannot be advertised as reaching the second scale. The finite initial-polynomial construction has no such restriction. Compare [remark 2.5](#).

Example 9.4 (A coefficient with nondiscrete support). Let $\gamma_k = 1 - 1/(k + 2)$ and $A = \sum_{k \geq 0} t^{\gamma_k}$. Its support is increasing and well ordered of order type ω , although all its exponents lie below 1 and there are infinitely many of them below 1. Since $v(A) = 1/2$, both roots of $P(z) = z^2 - A$ have valuation $1/4$; at that scale the reduction is $X^2 - 1$, so the residues are 1 and -1 . Writing $A = t^{1/2}(1 + B)$ with $B = \sum_{k \geq 1} t^{\gamma_k - 1/2}$ of positive well-ordered support, the two exact roots are

$$\alpha_{\pm} = \pm t^{1/4} \sum_{j \geq 0} \binom{1/2}{j} B^j. \quad (9.1)$$

The binomial series is strongly summable by [lemma 2.4](#) for the positive support $S = \{\gamma_k - 1/2 : k \geq 1\}$. Thus each root has support in $1/4 + S^*$; the shift comes from the displayed factor $t^{1/4}$. Its use does not require that the original coefficient series converge after replacing t by a positive real number $q \in (0, 1)$; in fact $q^{\gamma_k} \rightarrow q$, not 0, so the summands would fail to tend to zero. No such substitution is used in the Hahn construction, and the support is not an ordinary integer-power series in one monomial.

10 Polynomial image balls, finite maps, branch values, and ramification

The initial-polynomial identity determines a polynomial map on an entire valuation ball, not only its zero fiber. The *scale-local* statement of [theorem 10.1](#) and the *global algebraic* statement of [theorem 10.3](#) answer different questions and neither contains the other: the first is sharp on one prescribed ball and says nothing about the rest of K ; the second is exact on every fiber of the whole map and says nothing about scale. Both are kept.

10.1 Exact image balls and two mapping degrees

Theorem 10.1 (Exact image and fiber degree). *Let $P \in K[z]$ be nonconstant, write $P(a + Y) = b_0 + \sum_{j \geq 1} b_j Y^j$ and put*

$$\lambda = \min_{j \geq 1} \{v(b_j) + j\rho\}, \quad d_- = \min\{j \geq 1 : v(b_j) + j\rho = \lambda\}, \quad d_+ = \max\{j \geq 1 : v(b_j) + j\rho = \lambda\}.$$

Then

$$P(B_{\geq}(a, \rho)) = B_{\geq}(P(a), \lambda), \quad (10.1)$$

$$P(B_{>}(a, \rho)) = B_{>}(P(a), \lambda). \quad (10.2)$$

Every target in (10.1) has exactly d_+ preimages in the closed source ball, counted with polynomial multiplicity, and every target in (10.2) has exactly d_- preimages in the open source ball.

Proof. For $v(h) \geq \rho$ every term $b_j h^j$ with $j \geq 1$ has valuation at least λ , and for $v(h) > \rho$ strictly greater; this gives both inclusions. Let $v(y - P(a)) \geq \lambda$. The polynomial $t^{-\lambda}(P(a + t^\rho X) - y)$ has

finite coefficients and reduction of degree d_+ , since the changed constant term cannot affect the nonzero top coefficient; by [theorem 8.2](#) it has exactly d_+ roots in $\mathcal{O}_K$, proving surjectivity and the closed count. If $v(y - P(a)) > \lambda$, its reduced constant term vanishes and its order at zero is exactly d_- , and the open-ball count of the same theorem finishes the proof. $\square$

This is an arbitrary-rank polynomial counterpart of the usual non-Archimedean disk-mapping theorem; compare [\[19, Corollary 3.11\]](#). The finite proof shows why neither a complete real-valued norm nor a restricted-power-series convergence hypothesis is needed. The two mapping degrees genuinely differ: for $P(z) = z^2 - z$ at $a = 0$, $\rho = 0$ one has $\lambda = 0$, $d_- = 1$, $d_+ = 2$, so the finite elements map onto the finite elements with degree two while the infinitesimal ball at zero maps bijectively to itself, the second preimage of an infinitesimal target lying in the monad of 1. “Finite” here means exact fiber cardinality with multiplicity on a ball, not topological properness.

Corollary 10.2 (Affine normalization of a finite polynomial problem). *For finitely many nonzero polynomials of bounded finite degree one can choose one affine change of coordinates over K so that all their roots are finite in the rescaled coordinate; each nonzero polynomial can then be made monic with finite coefficients by a scalar normalization. Concretely, for $P = z^n + \sum_{j < n} a_j z^j$, $n \geq 1$, choose $\rho \in \Gamma$ with $\rho \leq \min_{a_j \neq 0} v(a_j)/(n - j)$; then $t^{-n\rho}P(t^\rho X)$ is monic with integral coefficients, and every root of a monic integral polynomial is integral.*

Proof. Choose a workspace containing all the roots and a ρ no larger than the finitely many valuations of the nonzero root displacements from the selected center; then every $(\alpha - a)/t^\rho$ is finite. Divide the rescaled polynomial by its leading coefficient and use Vieta. For the explicit criterion, the substitution multiplies the coefficient of z^j by $t^{(j-n)\rho}$, so integrality is exactly $v(a_j) + (j - n)\rho \geq 0$. Integrality of the roots is the argument after [theorem 3.1](#). Only a minimum among finitely many exponents is used. Constant nonzero polynomials have no roots to constrain the scale; if all relevant coefficients vanish, any scale works. The zero polynomial is excluded: its zero set is the whole field, which has elements of unbounded magnitude when $\Gamma \neq \{0\}$. $\square$

The normalization changes the numerical valuation precision of the coefficients, so precision must always be measured *after* the stated change of coordinates. This is what makes the integral-coefficient hypotheses of [section 13](#) harmless, and [section 13.3](#) records the exact bookkeeping.

10.2 Every fiber has the same algebraic length

Theorem 10.3 (A polynomial is a finite flat map of its degree). *Let $P \in K[z]$ be monic of degree $n \geq 1$ and regard $K[z]$ as a $K[Y]$ -algebra by $Y \mapsto P(z)$. It is free of rank n with basis $1, z, \dots, z^{n-1}$. For every $y \in K$ the fiber algebra $K[z]/(P(z) - y)$ has dimension n and decomposes as*

$$\prod_{P(\alpha)=y} K[u]/(u^{e_\alpha}), \quad e_\alpha = \text{ord}_\alpha(P - y), \quad \sum_{P(\alpha)=y} e_\alpha = n. \quad (10.3)$$

Proof. The presentation $K[Y, z]/(P(z) - Y) \cong K[z]$ by eliminating Y ; the defining polynomial is monic in z , so division makes the algebra free over $K[Y]$ with the stated basis, and specializing Y to y leaves the same basis and rank. [Theorems 3.1](#) and [3.2](#) give (10.3). $\square$

The words “finite flat” describe this free algebra and the associated map of *set-sized* spectra. They do not assert topological properness or path lifting in the full fine surreal topology; the ordinary connected real-parameter paths excluded in [\[4\]](#) play no role.

Theorem 10.4 (Branch-value polynomial). *Let $P \in K[z]$ be monic of degree $n \geq 2$ and list the roots of P' as $c_1, \dots, c_{n-1}$ with multiplicity. Then*

$$\text{Disc}_z(P(z) - Y) = (-1)^{n(n-1)/2} n^n \prod_{j=1}^{n-1} (P(c_j) - Y). \quad (10.4)$$

It has degree $n - 1$ in Y , and its zeros are exactly the values with a nonreduced fiber, that is, the critical values of P , with the multiplicities indicated by the product.

Proof. Put $A(z) = P(z) - Y$. The monic discriminant identity, the resultant symmetry sign $(-1)^{n(n-1)} = 1$, and the leading coefficient n of P' give

$$\text{Disc}_z A = (-1)^{n(n-1)/2} \text{Res}_z(A, P'), \quad \text{Res}_z(A, P') = \text{Res}_z(P', A) = n^n \prod_{j=1}^{n-1} A(c_j),$$

which together are (10.4). A fiber has a multiple root exactly when A and $A' = P'$ have a common root, so the zero set is as claimed; the leading coefficient in Y is nonzero in characteristic zero. $\square$

Coincident critical values combine their multiplicities in this product; there need not be $n - 1$ distinct branch values. The formula continues to make sense over coherent coefficient rings as a universal polynomial identity, even when the values cannot be labelled globally.

Proposition 10.5 (Finite and infinite ramification). *For a nonconstant degree- n polynomial over K , the local index at a finite point α is $e_\alpha = \text{ord}_\alpha(P - P(\alpha))$ and*

$$\sum_{\alpha \in K} (e_\alpha - 1) = n - 1, \quad (10.5)$$

only finitely many summands being nonzero. At infinity the local index is n , so the total sum on the algebraic projective line, including infinity, is $2n - 2$.

Proof. The finite Taylor expansion at α begins with a nonzero term of degree e_α , so the multiplicity of α in P' is $e_\alpha - 1$; summing the multiplicities of the degree- $(n - 1)$ polynomial P' gives (10.5). At infinity use $u = 1/z$, $w = 1/Y$: with leading coefficient a_n , $1/P(1/u) = a_n^{-1}u^n(1 + \dots)^{-1}$ has a unit factor with nonzero constant term, so the order is n ; adding $n - 1$ finishes. This is the polynomial instance of the ramification formula, derived directly rather than from a topological covering or surface argument. $\square$

Corollary 10.6 (Surjectivity and polynomial injections). *Every nonconstant polynomial $\mathbf{SC} \rightarrow \mathbf{SC}$ is surjective, and the injective ones are exactly the affine polynomials with nonzero slope.*

Proof. Algebraic closure supplies a root of $P - y$ for every target. If $n \geq 2$, choose y outside the finite set of critical values of theorem 10.4; its fiber has n distinct roots, so the map is not injective. The affine case is immediate. This proof is finite and does not require an analytic inverse theorem. $\square$

11 Critical points at every scale

The residue field is $\mathbb{C}$, so every nonzero integer is a unit by (2.3). That seemingly small fact makes the following counts exact, and they would fail in positive residue characteristic.

11.1 Differentiation commutes with nonconstant reduction

Lemma 11.1 (The derivative of an initial polynomial). *Over K , if $I = I_{a,\rho}(P)$ has degree $k \geq 1$ then*

$$w_{a,\rho}(P') = w_{a,\rho}(P) - \rho, \quad I_{a,\rho}(P') = I', \quad (11.1)$$

and more generally $w_{a,\rho}(P^{(r)}) = w_{a,\rho}(P) - r\rho$ and $I_{a,\rho}(P^{(r)}) = I^{(r)}$ for $1 \leq r \leq k$.

Proof. Differentiating $t^{-\lambda}P(a + t^\rho X)$ with respect to X gives $t^{\rho-\lambda}P'(a + t^\rho X)$, whose coefficients remain finite and whose reduction is I' . Since $\mathbb{C}$ has characteristic zero and I is nonconstant, $I' \neq 0$, so the displayed normalization has weighted valuation exactly zero, which is (11.1). For higher derivatives $I^{(r)} \neq 0$ whenever $r \leq k$ and the same argument applies. $\square$

The hypothesis cannot simply be removed: a constant initial polynomial has zero derivative and does not determine the next nonzero scale of P' . Likewise the argument can fail in positive residue characteristic, where a nonconstant reduction may have zero derivative.

Theorem 11.2 (Exact critical-point conservation in occupied balls). *Let $P \in K[z]$ be nonconstant and let B be either $B_{\geq}(a, \rho)$ or $B_{>}(a, \rho)$. If B contains exactly $k \geq 1$ roots of P counted with multiplicity, then for every $1 \leq r \leq k$ it contains exactly $k - r$ roots of $P^{(r)}$, counted with multiplicity; in particular exactly $k - 1$ critical points of P . Moreover, if $I = I_{a,\rho}(P)$ has positive degree, the residue directions and critical multiplicities inside the closed ball are read off from I' .*

Proof. For the closed ball, $\deg I = k$ by (8.5); lemma 11.1 gives an initial polynomial $I^{(r)}$ of degree $k - r$ for $P^{(r)}$, and theorem 8.2 converts that into the closed-ball count. For the open ball, $\text{ord}_0 I = k$ by (8.6), so $\deg I \geq k$ and the lemma still applies; in characteristic zero $\text{ord}_0 I^{(r)} = k - r$ in that range, which is the open-ball count. For the last assertion apply theorem 8.2 to P' , whose initial polynomial is exactly I' by lemma 11.1. $\square$

Remark 11.3 (A root-free ball is a different case). One must not substitute $k = 0$. The open infinitesimal ball at zero contains no root of $z^2 - 1$ but does contain the critical point zero. No formula with the meaningless count $k - 1 = -1$ is intended. Such critical points occur in residue directions *between* distinct root branches, which is exactly what theorem 11.6 makes explicit. In positive residue characteristic even the nonempty-ball argument can fail, because I' may lose its expected degree or vanish identically.

Corollary 11.4 (Ramification count on an arbitrary mapping ball). *With the notation of theorem 10.1, the closed source ball contains exactly $d_+ - 1$ critical multiplicities of P and the open source ball exactly $d_- - 1$. In either case, mapping degree one is equivalent to the absence of a critical point and implies bijectivity onto the image ball. The converse from bijectivity holds for closed balls, and for open balls provided $\Gamma \neq \{0\}$.*

Proof. Apply theorem 11.2 to $P - P(a)$, whose closed and open root counts are d_+ and d_- by theorem 10.1, and whose derivative is P' . Degree one supplies one simple preimage for every target. Conversely, if the degree exceeds one, choose a target that is not one of the finitely many critical values. The closed image ball contains $P(a) + ct^\lambda$ for all $c \in \mathbb{C}$, so is infinite. For an open image ball and $0 < \eta \in \Gamma$, the points $P(a) + t^{\lambda+n\eta}$, $n \geq 1$, are distinct points of that ball. In the asserted cases a noncritical target therefore exists; its degree-many preimages are simple and distinct, so the map is not injective. $\square$

The nontriviality condition cannot be dropped from the open-ball converse. For $\Gamma = \{0\}$, one has $K = \mathbb{C}$ and $B_{>}(0, 0) = \{0\}$. The restriction of $P(z) = z^2$ to this singleton is bijective onto its singleton image, although $d_- = 2$ and zero is a critical point. All the multiplicity counts above remain valid. Bijectivity of one exceptional fiber does not determine mapping degree.

Corollary 11.5 (Valuative Gauss–Lucas and nearest neighbours). *Every valuation ball containing all roots of P contains all its critical points. If $P \in K[z]$ is squarefree of degree at least two, then*

$$\max_{P'(w)=0} v(w - \alpha_i) = \delta_i = \max_{j \neq i} v(\alpha_i - \alpha_j). \quad (11.2)$$

Thus each root has a critical point at exactly its nearest-neighbour valuation, and no critical point is closer in the valuation sense.

Proof. The first assertion is [theorem 11.2](#) with $k = \deg P$. For the second, $B_{>}(\alpha_i, \delta_i)$ contains only α_i , whereas $B_{\geq}(\alpha_i, \delta_i)$ contains at least two roots; so the open ball contains no critical point and the closed ball contains at least one, and their difference is the shell $v(w - \alpha_i) = \delta_i$. $\square$

This belongs to the non-Archimedean root-location picture, which has precedents such as [\[20, 21\]](#). It is *not* a proof of a Euclidean root–critical-point distance conjecture: valuation distance ignores the ordinary leading coefficient of a modulus, and its balls are different objects from order disks. Our proof is a direct arbitrary-rank polynomial argument and uses no Berkovich space.

11.2 The branch polynomial between child clusters

Theorem 11.6 (Critical residue directions inside a cluster). *Suppose the closed ball $B_{\geq}(a, \rho)$ over K contains $k \geq 1$ roots and its nonempty root residue directions are the distinct $c_1, \dots, c_s \in \mathbb{C}$ with multiplicities $m_1, \dots, m_s$, $\sum_j m_j = k$, so that*

$$I_{a,\rho}(P) = C \prod_{j=1}^s (X - c_j)^{m_j}, \quad C \neq 0. \quad (11.3)$$

Then each occupied child ball $a + t^\rho(c_j + \mathfrak{m}_K)$ contains exactly $m_j - 1$ critical multiplicities, and the remaining critical points of the parent ball have total multiplicity $s - 1$, with residue directions given by the roots, with multiplicity, of the branch polynomial

$$W(X) = \sum_{j=1}^s m_j \prod_{\ell \neq j} (X - c_\ell). \quad (11.4)$$

It has degree $s - 1$, leading coefficient $k \neq 0$, and no root equal to any c_j .

Proof. Differentiating [\(11.3\)](#) gives $I' = C \prod_j (X - c_j)^{m_j-1} W(X)$. The leading coefficient of W is $\sum_j m_j = k \neq 0$, so $\deg W = s - 1$, and $W(c_j) = m_j \prod_{\ell \neq j} (c_j - c_\ell) \neq 0$. Now read every critical direction and multiplicity from I' by [lemma 11.1](#) and [theorem 11.2](#). Counting, $(k - 1) - \sum_j (m_j - 1) = s - 1$. $\square$

For two child directions the intervening critical direction is

$$\frac{m_1 c_2 + m_2 c_1}{m_1 + m_2},$$

with the *opposite* cluster multiplicities as weights; it is generally not the centroid of the roots. For $X^m(X-1)$ the nonzero critical direction is $m/(m+1)$ whereas the root centroid is $1/(m+1)$. More generally W is a weighted incomplete polynomial in the sense of [proposition 4.3](#), so its roots lie in the ordinary complex convex hull of the child directions. This is a useful bridge between the two geometries: the valuation locates the scale, and an ordinary complex polynomial locates the leading directions at that scale. The roots of W may themselves repeat and are counted with multiplicity, and W may be applied again to any smaller root-containing ball.

11.3 The finite root-cluster tree

For a finite set of distinct roots with positive multiplicities build a tree as follows. If more than one distinct root is present, take the full root set as the root node. For a node C containing at least two distinct roots set $\delta_C = \min_{\alpha \neq \beta \in C} v(\alpha - \beta)$; by the ultrametric inequality $v(\alpha - \beta) > \delta_C$ is an equivalence relation on C , and it has at least two classes since the minimum is attained. These classes are the occupied residue-direction children of C . Recurse until the children are singletons; a leaf is a single root location retaining its multiplicity. Recursion strictly reduces the number of distinct roots in a child, so the tree is finite even when the valuations have arbitrarily high ordered-group rank. Its depths belong to Γ , not necessarily to $\mathbb{R}$.

Every internal node C is exactly the set of roots in $B_{\geq}(a, \delta_C)$ for $a \in C$: a child lies in one residue subball of its parent and has strictly larger internal depth, while all roots outside it are separated at the parent's depth or earlier. Each internal node therefore has an actual scale and a branch polynomial ([11.4](#)).

Corollary 11.7 (Critical allocation by the root tree). *For a degree- n polynomial over K with d distinct roots, an internal node with s occupied children accounts for exactly $s-1$ critical multiplicities in the part of its ball outside those children, and a leaf of root multiplicity m accounts for $m-1$ at that root. These allocations account for all $n-1$ critical multiplicities, since*

$$\sum_{\text{internal nodes}} (s-1) = d-1, \quad \sum_{\text{leaves}} (m-1) = n-d.$$

Proof. At an internal node [theorem 11.6](#) gives $s-1$ outside its occupied children. A child carrying m total root multiplicities contains $m-1$ critical multiplicities, and its smallest closed root ball contains that same number by [theorem 11.2](#), so none are lost in the intermediate part of the child ball. At a singleton leaf the exact root multiplicity gives $m-1$ at the root. The tree identity follows by counting edges: the sum of all child counts equals the number of nodes minus one. If $d=1$ the direct multiplicity calculation already gives the assertion. The top ball contains all $n-1$ critical multiplicities, so the allocation is exhaustive. $\square$

This is an allocation theorem, not a claim that pairwise root valuations alone determine every critical point. The residue directions c_j carry additional data, and further subclustering among the roots of a branch polynomial can require its own rescaling analysis.

Example 11.8 (Critical points between nested roots). Let $t = \omega^{-1}$ and $P(z) = z(z-t^3)(z-t)(z-1)$, whose roots form the nested clusters $\{0, t^3\} \subset \{0, t^3, t\} \subset \{0, t^3, t, 1\}$.

ρ	$w_{0,\rho}(P)$	$I_{0,\rho}(P)$	$I_{0,\rho}(P')$
0	0	$X^3(X-1)$	$X^2(4X-3)$
1	3	$-X^2(X-1)$	$-X(3X-2)$
3	7	$X(X-1)$	$2X-1$

The three critical points are simple, with leading terms

$$w_{\text{outer}} = \frac{3}{4} + \text{infinitesimal}, \quad w_{\text{middle}} = t(\frac{2}{3} + \text{infinitesimal}), \quad w_{\text{inner}} = t^3(\frac{1}{2} + \text{infinitesimal}). \quad (11.5)$$

At scale zero two critical multiplicities remain in the zero monad and one has direction $3/4$; at scale one, one remains in the smaller zero child and one has direction $2/3$; at scale three the final critical point has direction $1/2$. The finite tree has three binary splits, each supplying one intervening critical point. Directly, $P'(z) = 4z^3 - 3(1+t+t^3)z^2 + 2(t+t^3+t^4)z - t^4$, whose coefficient profile verifies the critical valuations $0, 1, 3$; the branch calculation additionally supplies their leading complex coefficients and shows why all three are simple.

11.4 Two layer formulas for the discriminant valuation

The discriminant valuation of a monic polynomial with distinct finite roots can be decomposed either by *levels* or by *tree nodes*. The two decompositions are equivalent; each is convenient for a different purpose, and both are recorded.

Proposition 11.9 (Discriminant as a finite level sum). *Let $P \in K[z]$ be monic with distinct finite roots. List the distinct positive pairwise distance valuations as $0 < \lambda_1 < \dots < \lambda_r$, put $\lambda_0 = 0$, and for each k let $\mathcal{C}_k$ be the classes of the equivalence $\alpha_i \sim_k \alpha_j \iff v(\alpha_i - \alpha_j) \geq \lambda_k$. Then*

$$v(\text{Disc } P) = 2 \sum_{k=1}^r (\lambda_k - \lambda_{k-1}) \sum_{C \in \mathcal{C}_k} \binom{|C|}{2}. \quad (11.6)$$

If there are no positive pairwise valuations the sum is zero.

Proof. The ultrametric inequality makes $\sim_k$ an equivalence relation, and all pairwise distance valuations are nonnegative because the roots are finite. A pair of distance valuation λ_j lies in a common class at levels $k \leq j$ and not above, so its contribution before the factor 2 is $\sum_{k \leq j} (\lambda_k - \lambda_{k-1}) = \lambda_j$; a pair of valuation zero contributes nothing. Sum over the finitely many pairs and use (3.12). $\square$

Proposition 11.10 (Discriminant as a sum over the root tree). *Let $P \in K[z]$ be monic and squarefree of degree $n \geq 2$. With the tree of section 11.3,*

$$v(\text{Disc } P) = n(n-1)\delta_{\text{root}} + \sum_{\substack{C \text{ internal} \\ C \neq \text{root}}} |C|(|C|-1)(\delta_C - \delta_{\text{parent}(C)}). \quad (11.7)$$

The depths δ_C can be negative, positive, or non-Archimedean elements of Γ . Formulas (11.6) and (11.7) agree whenever both apply.

Proof. The separation valuation of a pair of distinct roots is the depth of its least common ancestor: start at the root depth and add depth increments for every proper internal node containing both roots, a telescoping sum. Summing over unordered pairs counts $\binom{|C|}{2}$ pairs at each node; multiplying by two and using (3.12) gives (11.7). For the comparison, both expressions compute $2 \sum_{i < j} v(\alpha_i - \alpha_j)$, the first by grouping the pairs according to which levels λ_k they survive and the second by grouping them according to their least common ancestor. When the roots are finite, $\delta_{\text{root}} \geq 0$ and the classes $\mathcal{C}_k$ are exactly the node sets of the tree cut at height λ_k . $\square$

Both formulas are meaningful in any ordered divisible value group. They are finite telescoping identities, not integrals with respect to a real-valued radius. They explain why discriminant precision can protect a whole nested root configuration, which is the content of corollary 13.4.

12 Support-controlled factor lifting

The preceding sections use finite algebra after the coefficients are given. Constructing factors from a perturbation may require infinitely many Hahn coefficients, and there the support convention becomes substantive. The argument is a coprime-factor version of the preparation recursions in [4, 6].

12.1 The ordinary finite linear splitting

Let $p_1, \dots, p_s \in \mathbb{C}[z]$ be pairwise coprime monic polynomials of degrees $d_i \geq 1$, $n = \sum_i d_i$, $P_0 = \prod_i p_i$. The map

$$L : \bigoplus_{i=1}^s \mathbb{C}[z]_{<d_i} \longrightarrow \mathbb{C}[z]_{<n}, \quad (h_i) \longmapsto \sum_i h_i \prod_{j \neq i} p_j \quad (12.1)$$

is an isomorphism: reducing a zero image modulo p_i forces $h_i = 0$, because the other factors are invertible modulo p_i , and source and target have the same finite dimension. This is the linearization of multiplication of monic factors, and the only linear inverse used below.

Theorem 12.1 (Support-controlled coprime Hensel factorization). *Let $P = P_0 + E \in \mathcal{O}_K[z]$ be monic of degree n with $\deg E < n$ and all coefficients of E infinitesimal, and let $S \subseteq \Gamma_{>0}$ be the union of their nonzero supports. There are unique monic $P_i = p_i + A_i$ with $\deg A_i < d_i$ and corrections of positive support such that*

$$P = \prod_{i=1}^s P_i. \quad (12.2)$$

All coefficients of all A_i have support in $S^ \setminus \{0\}$, and the P_i remain pairwise coprime over K . Divisibility of Γ is not required.*

Proof. Expanding the product, the equation reads $E = L(A_1, \dots, A_s) + N(A_1, \dots, A_s)$, where every monomial of N contains at least two correction factors. Proceed by well-founded recursion on $S^* \setminus \{0\}$. At exponent ν the nonlinear coefficient N_ν depends only on correction coefficients at exponents strictly smaller than ν , since the exponents of the two or more corrections sum to ν and each is positive; by lemma 2.4 there are finitely many contributions. Define

$$(A_{1,\nu}, \dots, A_{s,\nu}) = L^{-1}(E_\nu - N_\nu). \quad (12.3)$$

The degree bounds persist because both the linear splitting and every product difference in the monic product have degree less than n , and all factors remain monic of the prescribed degrees. The union of the constructed supports lies in the well-ordered S^* , and (12.3) verifies the product at every exponent.

For uniqueness, compare two positive-support factorizations and take the least exponent at which any correction differs, using the finite union of their well-ordered supports. At that exponent every product term involving an additional positive correction uses only smaller differences and vanishes, so the remaining equation is L applied to the vector of first differences equals zero; injectivity of L contradicts the choice of that exponent. This proves uniqueness among all positive-support candidates, not only those preassigned to S^* .

Finally the coefficient matrix of the finite Bézout map of a pair P_i, P_j reduces to the corresponding invertible matrix for p_i, p_j , so its determinant has nonzero standard part and is a unit of $\mathcal{O}_K$; the adjugate gives an inverse and hence a Bézout identity, so P_i, P_j are coprime. $\square$

No iteration through only the natural numbers is assumed cofinal in the output support: (12.3) is genuinely a recursion on a well-ordered support monoid, and remarks 2.5 and 2.6 are exactly what must be avoided. The certificate S^* is the universal support bound of this article.

Corollary 12.2 (Simple-root lifting and the first correction). *Every simple root c of P_0 lifts to a unique root $c + \eta$ of P with $\eta \in \mathfrak{m}_K$ and $\text{supp } \eta \subseteq S^* \setminus \{0\}$. If S is nonempty, at the least positive error exponent $e = \min S$ the coefficient is*

$$\eta_e = -\frac{E_e(c)}{P'_0(c)}. \quad (12.4)$$

There may be cancellation making this coefficient zero; no nonvanishing of the first correction is assumed. If $E = 0$, then $S = \emptyset$ and $\eta = 0$; there is no first error exponent.

Proof. Apply theorem 12.1 to the coprime pair $z - c$ and $P_0/(z - c)$, omitting the latter if it is 1; the lifted first factor is $z - c - \eta$, and for any root reducing to c , uniqueness of this linear factor identifies it with $c + \eta$. When $S \neq \emptyset$, expanding $P(c + \eta) = 0$ at its least exponent gives $P'_0(c)\eta_e + E_e(c) = 0$. $\square$

Corollary 12.3 (Canonical standard-part cluster factors). *For monic $P \in \mathcal{O}_K[z]$ write $\text{st}(P) = \prod_{c \in A} (z - c)^{m_c}$ with $A \subset \mathbb{C}$ finite. Then P has unique monic factors P_c with $\text{st}(P_c) = (z - c)^{m_c}$, and the roots of P_c are exactly the roots of P in $c + \mathfrak{m}_K$.*

Proof. Apply theorem 12.1 to the pairwise coprime ordinary factors. Every root of P_c is finite and its standard part is a root of $\text{st}(P_c)$ by (3.3), so all of them lie over c ; the product identity and degree counts show none are omitted. $\square$

Coprimality is essential for the support conclusion. Individual roots of a multiple cluster need not have supports in the original exponent group: for $P = z^m - t \in K_{\mathbb{Z}}[z]$ the reduction z^m has a repeated root, and the individual roots $\zeta t^{1/m}$ require a divisible workspace and lie outside the monoid generated by 1. The theorem controls the *coprime cluster factors*, not a nonexistent simple-root splitting of their common multiple ordinary reduction. Symmetric cluster coefficients can have substantially simpler supports than their separately labelled roots.

Example 12.4 (An explicit two-factor recursion). For $P = z^2 + tz - 1 + t^2$ the ordinary factors are $z - 1$ and $z + 1$, and the normalized roots have expansions

$$\alpha_+ = 1 - \frac{1}{2}t - \frac{3}{8}t^2 - \frac{9}{128}t^4 + O(t^6), \quad \alpha_- = -1 - \frac{1}{2}t + \frac{3}{8}t^2 + \frac{9}{128}t^4 + O(t^6),$$

obtainable either from (12.3) or from $\alpha_{\pm} = (-t \pm \sqrt{4 - 3t^2})/2$ with the strongly summable binomial series. The two factors are $z - \alpha_+$ and $z - \alpha_-$; their sum and product have the exact prescribed coefficients, and all displayed support lies in the positive integer monoid generated by the input.

12.2 Holomorphic parameters on one fixed ordinary domain

Theorem 12.5 (Parameter factor lifting without repeated shrinking). *Let $U \subseteq \mathbb{C}^d$ be one ordinary domain, fixed throughout, and let $A_1(x, z), \dots, A_s(x, z) \in \mathcal{O}(U)[z]$ be monic of fixed degrees d_i and pairwise coprime at every $x \in U$. Let $P = \prod_j A_j + E \in \mathcal{H}_{\Gamma, \geq 0}(U)[z]$ be monic of the same total degree n , with E of z -degree less than n and of strictly positive Hahn support with ordinary holomorphic coefficients on U , and let S be its total positive coefficient support. Then (12.2) holds*

with unique monic factors $A_j + B_j$, $B_j \in \mathcal{H}_{\Gamma, >0}(U)[z]$, $\deg B_j < d_j$, whose corrections have support in $S^* \setminus \{0\}$, and every ordinary coefficient function is holomorphic on the original domain U . No shrinking of U occurs at any step.

Proof. In the finite power bases, (12.1) is a square matrix $L(x)$ of ordinary holomorphic functions on U . Its value is invertible at every $x \in U$ by pairwise coprimality, so $L(x)^{-1} = \text{adj}(L(x))/\det L(x)$ is holomorphic on all of U ; there is no requirement that its ordinary size be uniformly bounded. Apply (12.3) coefficientwise with this fixed holomorphic matrix inverse. Finite products and applications of L^{-1} preserve holomorphy on the same U and introduce no Hahn exponent, so the support and uniqueness arguments are unchanged. $\square$

This theorem is stated over the fixed-domain ring $\mathcal{H}_{\Gamma, \geq 0}(U)$ and is proved only there. It is a finite-dimensional parameter version of the matrix inversion principle in [5]. It requires an ordinary coprime factorization on the given base; it does not manufacture globally labelled roots through ordinary monodromy or across a discriminant locus. When the leading ordinary polynomial has only simple roots, the same argument on its ordinary root cover gives a unique coherent lift of every sheet, and local lifts agree on overlaps by uniqueness, so ordinary analytic continuation permutes the lifted sheets exactly as it permutes the original ones. That does *not* produce a globally labelled root on a base with nontrivial ordinary monodromy, and it asserts nothing about holomorphic individual roots through a collision.

13 Perturbation: three theorems with three hypotheses

Three genuinely different perturbation statements are proved below and are deliberately *not* merged into one. They differ in hypothesis and in what they deliver.

Statement	Hypothesis on P	What it gives
theorem 13.1	none beyond monic integral; repeated roots allowed	a matching of root <i>multisets</i> with $v(\alpha_i - \beta_i) \geq \varepsilon/n$, and ε/n is optimal uniformly
theorem 13.3	distinct finite roots, and $\varepsilon > T(P) = \max_i(d_i + \delta_i)$	a <i>unique</i> labelling, the whole matrix $v(\beta_i - \beta_j) = d_{ij}$ with leading coefficients, exact displacement valuations, a second-order Newton error, $v(Q'(\beta_i)) = d_i$ and $v(\text{Disc } Q) = v(\text{Disc } P)$
corollary 13.4	distinct finite roots and $\varepsilon > v(\text{Disc } P)$	everything in theorem 13.3 , from a single discriminant valuation; it can require strictly more precision

It must be emphasized that [theorem 13.1](#) is *not* a weaker version of [theorem 13.3](#) and does not follow from it. It assumes no root separation whatsoever — in particular it applies to polynomials with repeated roots, where $T(P)$ is undefined — and its conclusion is correspondingly much weaker. The two have different hypotheses and both belong.

13.1 Optimal valuation-Hölder matching of root multisets

If $P, Q \in \mathcal{O}_K[z]$ are monic of degree n with coefficients agreeing to valuation at least ε , evaluating $Q - P$ at a root of P shows that *some* root of Q is close. That alone does not produce a bijection

of root multisets: several roots could be sent to the same nearby root. The next proof balances multiplicities in every relevant ultrametric cluster before matching.

Theorem 13.1 (Optimal valuation-Hölder root matching). *Let $P = z^n + \sum_{j < n} a_j z^j$ and $Q = z^n + \sum_{j < n} b_j z^j$ with $a_j, b_j \in \mathcal{O}_K$, $n \geq 1$, and suppose $v(a_j - b_j) \geq \varepsilon > 0$ for every $j < n$. Then there are listings of their roots, repeated with multiplicity, such that*

$$\boxed{v(\alpha_i - \beta_i) \geq \frac{\varepsilon}{n} \quad (1 \leq i \leq n).} \quad (13.1)$$

The exponent ε/n is optimal uniformly over all such polynomials.

Proof. All roots of both polynomials are finite. Put $\rho = \varepsilon/n$. In the finite combined root multiset, inspect the valuations of the nonzero pairwise differences that are strictly less than ρ , and choose μ with $0 < \mu < \rho$ greater than all of these finitely many valuations; such a μ exists by divisibility — take the midpoint between ρ and the maximum of that finite set together with zero.

On the finite collection of root locations the equivalence relations $v(x - y) \geq \mu$ and $v(x - y) \geq \rho$ now coincide, and their classes are intersections with closed μ -balls. Take any class and choose a root a in it, from either polynomial. Translation by the finite center a preserves the lower bound ε for the coefficients of $E = Q - P$, since each translated coefficient is a finite sum of an original coefficient times an integer and a nonnegative power of a , and (2.3) applies. Dilation by t^μ with $\mu > 0$ gives $w_{a,\mu}(E) \geq \varepsilon$. On the other hand (8.4) yields

$$w_{a,\mu}(P) = \sum_{i=1}^n \min\{\mu, v(\alpha_i - a)\} \leq n\mu < \varepsilon.$$

So corollary 8.3 says that this class contains exactly the same number of P -roots as Q -roots, with multiplicity; the same argument rules out a class containing roots of only one polynomial. Match the two finite multisets arbitrarily within each class. Its members have pairwise differences of valuation at least ρ , so every such pairing satisfies (13.1).

For optimality take $P = z^n$ and $Q = z^n - t^\varepsilon$: every root of Q has valuation exactly ε/n while every root of P is zero, so no larger uniform bound is possible. $\square$

The proof uses finitely many roots and their finite separation data. There is no passage to a limit of roots as an auxiliary radius increases, and it works when Γ has arbitrary ordered-group rank. The correspondence is not generally unique: repeated or unresolved clusters naturally allow several labellings.

Corollary 13.2 (Simultaneous stability for derivatives). *Under the hypotheses of theorem 13.1, for every integer $0 \leq r < n$ the root multisets of $P^{(r)}$ and $Q^{(r)}$ admit a matching with $v(\alpha_i^{(r)} - \beta_i^{(r)}) \geq \varepsilon/(n - r)$.*

Proof. Divide each r th derivative by the nonzero integer $n(n - 1) \cdots (n - r + 1)$, making it monic of degree $n - r$. Its coefficients remain finite and the coefficient differences retain valuation at least ε , because every nonzero rational factor has valuation zero by (2.3). Apply theorem 13.1. $\square$

For first derivatives the denominator is $n - 1$, not n , and it is sharp: $P = z^n$ and $Q = z^n - t^\varepsilon z$ change the normalized derivative from z^{n-1} to $z^{n-1} - t^\varepsilon/n$. The integrality hypothesis is used twice — roots are finite, and translating an error by a root does not reduce its coefficient valuation bound. For arbitrary coefficients, first apply corollary 10.2 to a common root scale and then measure the rescaled coefficient error; an unnormalized claim with the same ε/n at all infinite root locations would discard the conditioning factors introduced by that change of variables.

13.2 The rootwise threshold, and full cluster stability

Let $P \in \mathcal{O}_K[z]$ be monic of degree $n \geq 2$ with distinct finite roots $\alpha_1, \dots, \alpha_n$, and recall from [section 1.3](#)

$$d_{ij} = v(\alpha_i - \alpha_j), \quad d_i = v(P'(\alpha_i)) = \sum_{j \neq i} d_{ij}, \quad \delta_i = \max_{j \neq i} d_{ij}, \quad T(P) = \max_i (d_i + \delta_i). \quad (13.2)$$

All these are nonnegative. Here δ_i records the closest other root in valuation geometry — larger valuation means smaller distance — while d_i measures the size of the inverse derivative at the root. Distinguishing them gives a sharper threshold than any single coarse discriminant bound. The open balls $B_{>}(\alpha_i, \delta_i)$ are pairwise disjoint, since a point in two of them would force $v(\alpha_i - \alpha_j) > \min(\delta_i, \delta_j) \geq d_{ij}$.

The following theorem is proved, in the same form and with the same threshold, by all three merged manuscripts; two of them state it with the letters $T(P)$ and $\tau(P)$ for the same number $\max_i (d_i + \delta_i)$, and with σ_i and d_i for the same $v(P'(\alpha_i))$. It is stated once. Its conclusion list is the union of theirs: the unique labelling and the preserved distance matrix, the leading-coefficient refinement, the exact displacement valuation, and the second-order Newton estimate [\(13.9\)](#), which only one of the three proves.

Theorem 13.3 (Exact matching, full cluster stability, and a second-order Newton error). *Let $P \in \mathcal{O}_K[z]$ be monic of degree $n \geq 2$ with distinct finite roots and data [\(13.2\)](#), and let $Q = P + E$ be monic of degree n with $\deg E < n$ and coefficient precision $\varepsilon = \min_{j < n} v([z^j]E)$, with $\varepsilon = +\infty$ when $E = 0$. Suppose*

$$\varepsilon > T(P) = \max_i (d_i + \delta_i). \quad (13.3)$$

Then Q is squarefree with n distinct finite roots, and there is a unique labelling $\beta_1, \dots, \beta_n$ with

$$v(\beta_i - \alpha_i) > \delta_i \quad (1 \leq i \leq n), \quad (13.4)$$

that is, exactly one root of Q in each disjoint ball $B_{>}(\alpha_i, \delta_i)$, and these are all of them. Moreover

$$v(\beta_i - \beta_j) = d_{ij} \quad \text{and} \quad \text{lc}(\beta_i - \beta_j) = \text{lc}(\alpha_i - \alpha_j) \quad (i \neq j), \quad (13.5)$$

$$v(\beta_i - \alpha_i) \geq \varepsilon - d_i, \quad (13.6)$$

$$v(Q'(\beta_i)) = d_i, \quad v(\text{Disc } Q) = v(\text{Disc } P). \quad (13.7)$$

If $E(\alpha_i) = 0$ then $\beta_i = \alpha_i$. If $E(\alpha_i) \neq 0$ then the displacement has the exact valuation and leading term

$$v(\beta_i - \alpha_i) = v(E(\alpha_i)) - d_i, \quad \beta_i - \alpha_i = -\frac{E(\alpha_i)}{P'(\alpha_i)}(1 + \epsilon_i), \quad \epsilon_i \in \mathfrak{m}_K, \quad (13.8)$$

and the first Newton correction is a genuine second-order improvement:

$$v\left(\beta_i - \alpha_i + \frac{E(\alpha_i)}{P'(\alpha_i)}\right) \geq 2(\varepsilon - d_i) - \delta_i. \quad (13.9)$$

Proof. Fix i and work at center α_i and scale δ_i . Formula [\(8.4\)](#) gives

$$w_{\alpha_i, \delta_i}(P) = \delta_i + \sum_{j \neq i} d_{ij} = \delta_i + d_i, \quad (13.10)$$

since every other separation has $d_{ij} \leq \delta_i$; the initial polynomial therefore has a simple zero at the origin, so exactly one root of P lies in $B_{>}(\alpha_i, \delta_i)$. Every coefficient of the translated expansion

$E(\alpha_i + Z)$ has valuation at least ε : use the finite binomial formula, integrality of α_i , and (2.3); since $\delta_i \geq 0$, scaling by t^{δ_i} cannot lower these valuations. Hence

$$w_{\alpha_i, \delta_i}(E) \geq \varepsilon > d_i + \delta_i = w_{\alpha_i, \delta_i}(P),$$

and [corollary 8.3](#) gives exactly one root of Q , counted with multiplicity, in $B_{>}(\alpha_i, \delta_i)$; it is therefore simple. These n balls are disjoint, their n simple roots exhaust the degree of Q , and finiteness of the roots is immediate from finiteness of the centers and nonnegativity of the scales. This proves the matching and its uniqueness.

Put $u_i = \beta_i - \alpha_i$. For $i \neq j$ both u_i, u_j have valuation strictly greater than d_{ij} , so $\alpha_i - \alpha_j$ determines both the valuation and the leading coefficient of $\beta_i - \beta_j$; this is (13.5). Taking valuations of the finite root products gives (13.7); alternatively $Q'(\alpha_i + u_i) = U(\alpha_i + u_i) + u_i U'(\alpha_i + u_i) + E'(\alpha_i + u_i)$ with $U = \prod_{j \neq i} (z - \alpha_j)$, where $u_i U' / U = \sum_{j \neq i} u_i / (\beta_i - \alpha_j)$ is infinitesimal and the last term has valuation at least $\varepsilon > d_i$.

For the displacement, all factors $\beta_i - \alpha_j$ with $j \neq i$ have valuation d_{ij} , so $v(P(\beta_i)) = v(u_i) + d_i$ when $u_i \neq 0$; since $P(\beta_i) = -E(\beta_i)$ and β_i is integral, $v(E(\beta_i)) \geq \varepsilon$, which is (13.6) (a zero displacement satisfies it under the $+\infty$ convention). If $E(\alpha_i) = 0$ then α_i itself is a root of Q in its assigned ball and uniqueness gives $u_i = 0$. Otherwise the exact factorization

$$P(\alpha_i + u_i) = P'(\alpha_i) u_i \prod_{j \neq i} \left(1 + \frac{u_i}{\alpha_i - \alpha_j} \right) = P'(\alpha_i) u_i (1 + \epsilon_i), \quad \epsilon_i \in \mathfrak{m}_K, \quad (13.11)$$

has every parenthesized factor a unit with standard part one, because $v(u_i) > \delta_i \geq d_{ij}$. Also $E(\alpha_i + u_i) - E(\alpha_i) = u_i V_i$ with $v(V_i) \geq \varepsilon$, by the finite difference-of-powers identity and integrality of both arguments; since $\varepsilon > d_i$ this difference has valuation strictly larger than $d_i + v(u_i)$ and cannot affect the leading term of $P(\alpha_i + u_i) + E(\alpha_i + u_i) = 0$. This proves (13.8).

For (13.9) set $R_P = P(\alpha_i + u_i) - P'(\alpha_i) u_i$ and $R_E = E(\alpha_i + u_i) - E(\alpha_i)$. Expanding the finite product in (13.11) and the finite Taylor polynomial of E gives

$$v(R_P) \geq d_i + 2v(u_i) - \delta_i, \quad v(R_E) \geq \varepsilon + v(u_i), \quad (13.12)$$

the first because every term of the product minus one contains at least one factor $u_i / (\alpha_i - \alpha_j)$ of valuation at least $v(u_i) - \delta_i$, the second because every translated error coefficient has valuation at least ε and $v(u_i) > 0$. The root equation is $P'(\alpha_i) u_i + E(\alpha_i) + R_P + R_E = 0$; dividing by $P'(\alpha_i)$ and inserting (13.12) and then (13.6) gives

$$v \left(u_i + \frac{E(\alpha_i)}{P'(\alpha_i)} \right) \geq \min \{ 2v(u_i) - \delta_i, \varepsilon + v(u_i) - d_i \} \geq 2(\varepsilon - d_i) - \delta_i. \quad \square$$

The theorem preserves the entire matrix of pairwise valuations, so every root partition defined by a valuation threshold, and hence the whole cluster tree of [section 11.3](#), is preserved. That is a precise meaning of preservation of the complete finite cluster geometry. It does *not* assert equality of the roots, of their ordinary moduli, or of every Hahn coefficient of the pairwise differences.

When $E \neq 0$, the right side of (13.9) is strictly greater than $\varepsilon - d_i$ under (13.3), so $-E(\alpha_i) / P'(\alpha_i)$ is a genuine first-order correction with a strictly smaller error, not merely a formal suggestion to apply Newton's method. No infinite Newton iteration is required for the proof. For degree one the exact root displacement is simply the negative constant-coefficient error.

Corollary 13.4 (Discriminant precision protects all clusters). *Under the hypotheses on P , put $D = v(\text{Disc } P)$. Then $T(P) \leq D$, so a monic same-degree perturbation with*

$$\varepsilon > D \tag{13.13}$$

has all the matching and preservation properties of [theorem 13.3](#). One also has $T(P) \leq 2 \max_i d_i$.

Proof. By (3.12), $D = 2 \sum_{i < j} d_{ij} = \sum_i d_i$. Fix i and choose j_0 with $d_{ij_0} = \delta_i$; in $d_i + \delta_i$ the pair (i, j_0) is counted twice, every other pair incident to i once, and the remaining pairs not at all, and all d_{ij} are nonnegative, so $d_i + \delta_i \leq D$. Take the maximum over i . For the last bound, $\delta_i \leq d_i$ since all separations are nonnegative. $\square$

Example 13.5 (Why the inequality must be strict). Let $h > 0$ be a value-group element and put

$$P(z) = z(z - t^h), \quad Q(z) = P(z) + \frac{1}{4}t^{2h} = (z - \frac{1}{2}t^h)^2.$$

The two roots of P are distinct with $d_i = \delta_i = h$, so $T(P) = D = 2h$. The error has valuation exactly $2h$, yet Q has a double root, which lies in neither separated open ball $B_{>}(0, h)$ nor $B_{>}(t^h, h)$. Thus replacing $>$ by $\geq$ in (13.3) or in (13.13) is false. The example proves sharpness of the strict boundary in this family; it does not claim that D is the optimal threshold for every polynomial, nor optimality of the bound for every individual perturbation. It also realizes equality in the global Hölder bound (13.1) for $n = 2$.

Example 13.6 (A complete four-root hierarchy, with $T(P) = 7 < D = 10$). Let $t = \omega^{-1}$ and

$$P(z) = (z - t)(z - t - t^3)(z + t)(z - 1), \tag{13.14}$$

with roots labelled in that order. The positive distance valuations are $d_{12} = 3$, $d_{13} = d_{23} = 1$, $d_{14} = d_{24} = d_{34} = 0$, so

$$(d_1, d_2, d_3, d_4) = (4, 4, 2, 0), \quad (\delta_1, \delta_2, \delta_3, \delta_4) = (3, 3, 1, 0), \tag{13.15}$$

giving $T(P) = 7$ whereas $D = 10$; the discriminant has leading term $16t^{10}$, and (11.6) returns the same valuation as $2(1 \cdot \binom{3}{2} + (3 - 1)\binom{2}{2}) = 10$. The discriminant threshold is therefore *not* optimal.

The critical points can be placed without solving a cubic. At scale zero about zero the initial polynomial is $X^3(X - 1)$, whose derivative is $X^2(4X - 3)$, so two critical points lie in the infinitesimal monad of zero and one has standard part $3/4$. At scale one about zero the initial polynomial is, up to a nonzero scalar, $(X - 1)^2(X + 1)$, whose derivative is $(X - 1)(3X + 1)$: one critical point has $w/t \in 1 + \mathfrak{m}_K$ and another has $w/t \in -\frac{1}{3} + \mathfrak{m}_K$. In the ball of scale three about t the initial polynomial is a nonzero scalar times $X(X - 1)$, so the first of these has $(w - t)/t^3 \in \frac{1}{2} + \mathfrak{m}_K$. These three exhaust $\deg P' = 3$.

Now take $\varepsilon = 8$, between $T(P)$ and D . For $Q = P + t^8$, [theorem 13.3](#) applies and preserves every pairwise root distance valuation, with

$$\begin{aligned} \beta_1 - \alpha_1 &= -\frac{1}{2}t^4(1 + \epsilon_1), & \beta_2 - \alpha_2 &= \frac{1}{2}t^4(1 + \epsilon_2), \\ \beta_3 - \alpha_3 &= \frac{1}{4}t^6(1 + \epsilon_3), & \beta_4 - \alpha_4 &= -t^8(1 + \epsilon_4), \end{aligned}$$

all $\epsilon_i \in \mathfrak{m}_K$, directly from $-t^8/P'(\alpha_i)$. For $Q = P + t^8(1 + z)$ the four matched displacement valuations are exactly 4, 4, 6, 8 by (13.8). The original cluster scales 0, 1, 3 and this quartic's residual critical directions $3/4, -1/3, 1/2$ survive such a perturbation. The analogous statement for the different quartic of [example 11.8](#) preserves its own directions $3/4, 2/3, 1/2$.

Corollary 13.7 (Preservation of the tree and of its critical directions). *Under [theorem 13.3](#), at every internal node of the original root-cluster tree, using its original center and scale, P and Q have the same initial polynomial. Their critical multiplicities in every residue direction of that node therefore agree as well.*

Proof. A node may be centered at one of its roots α_i , and its depth ρ is at most δ_i ; all roots are finite so $\rho \geq 0$. By [\(8.4\)](#),

$$w_{\alpha_i, \rho}(P) = \rho + \sum_{j \neq i} \min\{\rho, d_{ij}\} \leq \delta_i + d_i < \varepsilon \leq w_{\alpha_i, \rho}(E),$$

so [corollary 8.3](#) gives equality of the initial polynomials. At an internal node this polynomial is nonconstant, so [lemma 11.1](#) gives equality for the derivatives too, and the direction counts follow from [theorems 8.2](#) and [11.6](#). $\square$

This preserves the root-scale hierarchy and the *allocated* critical directions, not the individual multiplicities of all critical points. For instance $z^3 - 1$ is squarefree but its derivative has a double root at zero; adding $t^N z$ with $N > 0$ leaves the ordinary root clusters unchanged and splits that critical point into two at a smaller scale, while the direction count at the original scale is still two.

13.3 Using the theorem at infinite scales

Finiteness of the roots in [theorem 13.3](#) is a normalization hypothesis, not a restriction to an Archimedean problem. Let P be any monic squarefree degree- n polynomial over K . Choose $a \in \mathbf{SC}$ and a positive surreal ρ_0 large enough that all $(\alpha_i - a)/\rho_0$ are finite, which [proposition 4.1](#) supplies from the coefficients, and put

$$P_*(Y) = \rho_0^{-n} P(a + \rho_0 Y), \quad Q_*(Y) = \rho_0^{-n} Q(a + \rho_0 Y),$$

both monic of degree n . If $E(a + Z) = \sum_{j < n} \tilde{e}_j Z^j$, the normalized precision is

$$\varepsilon_* = \min_{j < n} (v(\tilde{e}_j) + (j - n)v(\rho_0)), \quad (13.16)$$

and the discriminant transforms exactly as

$$\text{Disc}(P_*) = \rho_0^{-n(n-1)} \text{Disc}(P), \quad D_* = v(\text{Disc } P) - n(n-1)v(\rho_0). \quad (13.17)$$

Thus $\varepsilon_* > D_*$ is a coefficient-level sufficient condition for the normalized matching, and pulling the roots back by $a + \rho_0 Y$ preserves all pairwise valuation equalities in the original coordinates. Applying the unnormalized discriminant criterion without this adjustment would be unjustified when some original roots are infinite.

14 Universal residue duality and collision-stable traces

This section deliberately works over a more general coefficient ring than K . Let R be *any* commutative ring with identity and

$$P(z) = z^n + p_{n-1}z^{n-1} + \cdots + p_0 \in R[z], \quad n \geq 1.$$

Monic division works over R without dividing by any coefficient, so $A = R[z]/(P)$ is free with basis $1, z, \dots, z^{n-1}$, and we define

$$\lambda_P : A \longrightarrow R, \quad \lambda_P(h) = [z^{n-1}] \operatorname{rem}_P h, \quad (14.1)$$

the coefficient of z^{n-1} in the remainder of h on division by P . For $h \in A$ write m_h for its multiplication endomorphism. No contour is chosen and no moving root is labelled. In particular, if $P, h \in \mathcal{O}_K[z]$ then $\operatorname{rem}_P h$ and $\lambda_P(h)$ are integral as well, since monic division uses no inverse coefficients. The functional detects nilpotent directions that the trace does not, and it is defined by finite polynomial division, even at repeated roots and at infinite surcomplex roots.

14.1 A dual basis with no discriminant denominator

Theorem 14.1 (Universal residue pairing and explicit dual basis). *Over any commutative ring R with identity, the bilinear form $(f, g) \mapsto \lambda_P(fg)$ on $A = R[z]/(P)$ is perfect, including when P has repeated roots. With $p_n = 1$, the residue-dual basis to $1, z, \dots, z^{n-1}$ is*

$$b^i(z) = \sum_{k=i+1}^n p_k z^{k-1-i}, \quad 0 \leq i < n, \quad (14.2)$$

so that $\lambda_P(z^j b^i) = \delta_{ij}$. Its Gram matrix $G_{ij} = \lambda_P(z^{i+j})$ has

$$\det G = (-1)^{n(n-1)/2}, \quad (14.3)$$

and the coefficient matrix of the Bézoutian

$$\Delta_P(z, Y) = \frac{P(z) - P(Y)}{z - Y} = \sum_{i=0}^{n-1} b^i(z) Y^i \quad (14.4)$$

is G^{-1} in these power bases. The same pairing is perfect over $\mathcal{O}_K$ on $\mathcal{O}_K[z]/(P)$ when P is integral and monic.

Proof. There is a formal Laurent expansion at infinity, $P(z)^{-1} = z^{-n}(1 + p_{n-1}z^{-1} + \dots + p_0z^{-n})^{-1}$, the inverse being a formal power series in the independent symbol z^{-1} , valid over any R . Division by P gives

$$\lambda_P(h) = [z^{-1}] \frac{h(z)}{P(z)}, \quad (14.5)$$

since the quotient is polynomial and has no z^{-1} term, and for a remainder of degree less than n that coefficient is exactly its z^{n-1} coefficient. The definition of b^i yields

$$\frac{b^i(z)}{P(z)} = z^{-i-1} - \frac{\sum_{k=0}^i p_k z^{k-i-1}}{P(z)},$$

whose second term has highest possible Laurent degree $-n-1$; after multiplication by z^j with $0 \leq j < n$ it cannot contribute to degree -1 . So (14.5) gives $\lambda_P(z^j b^i) = \delta_{ij}$, constructing an inverse to the map $A \rightarrow \operatorname{Hom}_R(A, R)$ defined by the pairing and hence proving perfection over R itself.

For the determinant, $G_{ij} = 0$ when $i + j < n - 1$ and $G_{ij} = 1$ when $i + j = n - 1$; reversing the column order gives a lower triangular matrix with diagonal ones, and the column-reversal sign is $(-1)^{n(n-1)/2}$. Finally the finite identity $(z^k - Y^k)/(z - Y) = \sum_{i < k} z^{k-1-i} Y^i$ proves (14.4), whose i th coefficient column is the dual vector b^i , so its coefficient matrix is G^{-1} . $\square$

No inverse root separation, resultant or discriminant appears. This is stronger than nondegeneracy over a fraction field: it is an isomorphism over the given ring R , even when R has zero divisors or is non-Noetherian. Its strength for the present problem is that no discriminant denominator ever appears, which is what makes the pairing survive every collision. This is the polynomial core of the one-variable residue duality in [5]; it is not claimed as a new general duality theorem.

14.2 Trace, norm, and the derivative as the trace element

Theorem 14.2 (Universal trace identity). *Over any commutative ring R with identity and monic P of degree $n \geq 1$,*

$$\mathrm{Tr}(m_h) = \lambda_P(P'h) \quad (h \in A). \quad (14.6)$$

If $G_{\mathrm{tr},ij} = \mathrm{Tr}(m_{z^{i+j}})$ then

$$G_{\mathrm{tr}} = G M_{P'}, \quad \det G_{\mathrm{tr}} = \mathrm{Disc}(P), \quad (14.7)$$

where $M_{P'}$ is the multiplication matrix of P' and Disc is the usual universal coefficient polynomial for monic P . For any polynomial Q , $\det(m_Q) = \mathrm{Res}(P, Q)$.

Proof. The coefficient of z^i in the power-basis expansion of hz^i is $\lambda_P(hz^i b^i)$, so summing the diagonal coefficients gives $\mathrm{Tr}(m_h) = \lambda_P(h \sum_{i < n} z^i b^i)$. The explicit dual basis satisfies the polynomial identity

$$\sum_{i=0}^{n-1} z^i b^i = \sum_{k=1}^n \sum_{i=0}^{k-1} p_k z^{k-1} = \sum_{k=1}^n k p_k z^{k-1} = P', \quad (14.8)$$

proving (14.6) over every R . The trace-pairing entry is $\lambda_P(z^i P' z^j)$, the (i, j) entry of $G M_{P'}$. For $\det(m_Q) = \mathrm{Res}(P, Q)$, both sides are polynomials with integer coefficients in the finitely many coefficients of P, Q ; over the fraction field of the universal integer coefficient ring P has distinct roots and the multiplication determinant is $\prod_i Q(\alpha_i)$, which is (3.9). The universal coefficient ring injects into that field, so the identity holds there and after every specialization. Taking determinants in (14.7) gives $(-1)^{n(n-1)/2} \mathrm{Res}(P, P') = \mathrm{Disc}(P)$ for monic P , and the same universal-polynomial argument identifies it with (3.10) over splitting fields. $\square$

The element representing the trace under residue duality is exactly P' . This should not be mistaken for a proof of the general multivariable Jacobian comparison; the argument here concerns the monogenic algebra $R[z]/(P)$. The several-variable statement is theorem 17.2, proved separately.

Remark 14.3 (The Bézoutian is the duality kernel). There is a second, basis-free proof of (14.6) that will be reused verbatim in n variables. The element $\Delta_P \in A \otimes_R A$ of (14.4) satisfies $(z - Y)\Delta_P = 0$ and $(\lambda_P \otimes \mathrm{id})(\Delta_P) = 1$, the latter because its coefficient of z^{n-1} is one and its z -degree is $n - 1$. A perfect pairing identifies $A \otimes_R A$ with $\mathrm{End}_R(A)$ by

$$\sum_s u_s \otimes v_s \longmapsto \left(h \mapsto \sum_s \lambda_P(h u_s) v_s \right). \quad (14.9)$$

The first identity says the operator attached to Δ_P commutes with multiplication by z , hence with multiplication by every polynomial; the second says it sends 1 to 1. It is therefore the identity operator, so for residue-dual bases $(b_j), (b^j)$ one gets $\Delta_P = \sum_j b_j(z) b^j(Y)$ and $P' = \sum_j b_j b^j$ in A , and taking diagonal coefficients of multiplication by h recovers $\mathrm{Tr}(m_h) = \lambda_P(P'h)$. The derivative is the diagonal value of the duality kernel.

14.3 Residues at distinct or colliding roots

For a rational function, the residue at ξ means the coefficient of $(z - \xi)^{-1}$ in its formal Laurent expansion at ξ ; the residue at infinity is taken in the coordinate $u = 1/z$ for the differential, hence is the negative of the coefficient of z^{-1} in the expansion at infinity. These are formal Laurent coefficients, never integrals.

Proposition 14.4 (Finite residue formula at multiple roots). *Let P be monic over a characteristic-zero algebraically closed field, in particular over K , with $P = \prod_i (z - \alpha_i)^{m_i}$ and $P_i = P/(z - \alpha_i)^{m_i}$. Then for every polynomial h ,*

$$\lambda_P(h) = \sum_i \operatorname{Res}_{z=\alpha_i} \frac{h(z)}{P(z)} dz = \sum_i \frac{1}{(m_i - 1)!} \left(\frac{d^{m_i-1}}{dz^{m_i-1}} \frac{h(z)}{P_i(z)} \right)_{z=\alpha_i}, \quad (14.10)$$

and in the squarefree case

$$\lambda_P(h) = \sum_{i=1}^n \frac{h(\alpha_i)}{P'(\alpha_i)}. \quad (14.11)$$

Proof. Finite partial fractions express h/P as a polynomial plus a sum of terms $c_{i,k}(z - \alpha_i)^{-k}$. In the Laurent expansion at infinity only the terms with $k = 1$ contribute to z^{-1} , with contributions $c_{i,1}$; by (14.5) this proves the first equality. Near α_i , with $u = z - \alpha_i$, the coefficient $c_{i,1}$ is the coefficient of u^{m_i-1} in the Taylor expansion of $h(\alpha_i + u)/P_i(\alpha_i + u)$, whose denominator has nonzero constant term; that coefficient is exactly the displayed derivative. For a simple root $P_i(\alpha_i) = P'(\alpha_i)$. $\square$

Both formulas are finite algebraic identities for surcomplex inputs. The local terms in (14.11) may individually be infinite even when $\lambda_P(h)$ is finite or zero: this is not divergence but cancellation in a finite sum. For $P = z(z - t)$ one has $\lambda_P(h) = (h(t) - h(0))/t$, with infinite individual weights but with $\lambda_P(h)$ integral for $h \in \mathcal{O}_K[z]$, by (14.1). At the repeated polynomial $P = (z - t)^2$ one has $\lambda_P(h) = h'(t)$ — an exact formula involving a jet, not a limit taken as two roots approach each other. At a collision the separate point weights become singular while the universal functional remains defined and its local formula retains derivative information. The same partial-fraction argument gives the algebraic residue theorem for every rational differential on the projective line: the sum of its finitely many finite residues and its residue at infinity is zero. Again, no topological loop in **SC** is required.

Proposition 14.5 (Field-level traces and norms, including collisions). *Over an algebraically closed characteristic-zero field, with $P = \prod_i (z - \alpha_i)^{m_i}$,*

$$\operatorname{Tr}(m_h) = \sum_i m_i h(\alpha_i), \quad \operatorname{Nm}_A(h) := \det(m_h) = \prod_i h(\alpha_i)^{m_i}. \quad (14.12)$$

The radical of the trace pairing is the nilradical of A , whereas the residue pairing of theorem 14.1 has zero radical, whether or not roots repeat.

Proof. Use the CRT factors $K[u]/(u^{m_i})$ of theorem 3.2. On the i th factor multiplication by $h(\alpha_i + u)$ is triangular in the powers of u with every diagonal entry $h(\alpha_i)$, so its trace is $m_i h(\alpha_i)$ and its determinant $h(\alpha_i)^{m_i}$; summing and multiplying gives (14.12). On a pair of elements the trace pairing is $\sum_i m_i f(\alpha_i)g(\alpha_i)$; if all $f(\alpha_i) = 0$ this vanishes for every g , and conversely the CRT idempotent of the i th factor gives $m_i f(\alpha_i) = 0$, whence $f(\alpha_i) = 0$ in characteristic zero. These are precisely the elements whose image in each local factor is nilpotent. The residue pairing is perfect by its constructed dual basis. $\square$

Theorem 14.6 (Discriminant and trace-pairing degeneration). *For monic P of degree n over K with $T_{ij} = \text{Tr}(m_{z^{i+j}})$,*

$$\det T = \text{Disc}(P) = (-1)^{n(n-1)/2} \text{Res}(P, P'). \quad (14.13)$$

The trace pairing is nondegenerate exactly when P is squarefree, whereas the residue pairing of theorem 14.1 is always nondegenerate, with the constant Gram determinant (14.3).

Proof. List the n roots with repetition and let $V_{ki} = \alpha_k^i$. The trace formula gives $T = V^T V$ (transpose, not conjugate transpose), so $\det T = (\det V)^2 = \prod_{i < j} (\alpha_i - \alpha_j)^2$, an identity that remains valid with repeated rows and so handles repeated roots directly. Also $\prod_i P'(\alpha_i) = (-1)^{n(n-1)/2} \prod_{i < j} (\alpha_i - \alpha_j)^2$: in the squarefree case multiply $P'(\alpha_i) = \prod_{j \neq i} (\alpha_i - \alpha_j)$, and with repeated roots both sides vanish. Apply (3.9). The product of squared differences is nonzero precisely for distinct roots, while the residue determinant is the fixed unit (14.3). $\square$

This determinant formula shows that the product definition (3.10) is a coefficient polynomial, and links its valuation to the root-separation precision of section 13. The trace cannot detect nilpotent directions in a local factor, whereas the residue functional detects the higher jets needed for a perfect pairing; these are different bilinear forms and must not be interchanged. Compare also theorem 6.4, which is a third form again, built from the trace and used for real-root counting.

Example 14.7 (A cubic collision). This calculation appears in the supplied several-variable manuscript [5]; it is included as a normalization check and a bridge to that work, not as a new example. Let $P = z^3 - az - b$. In the basis $(1, z, z^2)$,

$$G = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & a \end{pmatrix}, \quad G^{-1} = \begin{pmatrix} -a & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \quad G_{\text{tr}} = \begin{pmatrix} 3 & 0 & 2a \\ 0 & 2a & 3b \\ 2a & 3b & 2a^2 \end{pmatrix},$$

with $\det G = -1$ and $\det G_{\text{tr}} = 4a^3 - 27b^2$. The residue-dual basis is $(z^2 - a, z, 1)$, whose trace element is $(z^2 - a) + z^2 + z^2 = 3z^2 - a = P'$. At $a = 3t^2$, $b = 2t^3$ the polynomial is $(z - 2t)(z + t)^2$ and

$$\lambda_P(h) = \frac{h(2t) - h(-t)}{9t^2} - \frac{h'(-t)}{3t}, \quad \text{Tr}(m_h) = h(2t) + 2h(-t). \quad (14.14)$$

The trace forgets the nilpotent direction at the double point; the residue functional remembers its first derivative.

14.4 Coherent coefficient families on a fixed ordinary domain

Take $R = \mathcal{H}_\Gamma(U)$ or $R = \mathcal{H}_{\Gamma, \geq 0}(U)$ for one fixed ordinary domain U . Every monic bounded-degree family $P(x, z)$ gives a free rank- n algebra over R . Polynomial division, multiplication matrices, λ_P , the dual basis (14.2), traces, norms and discriminants are computed by finitely many coefficient-ring operations, hence are legitimate Hahn-coherent coefficient functions on the *same* U . No inverse discriminant and no repeated domain shrink is needed for the residue pairing. For an integral family all entries of the residue Gram matrix and of its explicit inverse are integral, and if the input coefficient supports lie in a nonnegative support monoid M , so do these entries. At a finite surcomplex parameter, evaluation (2.4) is a ring homomorphism, so it commutes with these finite identities and the specialized pairing is exactly the pairing of the specialized polynomial. This explains collision stability algebraically, independently of any simultaneous choice of root labels.

For more general coherent analytic numerators of unbounded Taylor degree one must first use the fixed-domain division theorems of the supplied manuscripts before taking a remainder. The

purely polynomial argument of this section does not silently replace that extra analytic step, and nothing here is asserted over a radius-free or formal-coefficient ring.

15 Several variables: certificates, finite algebras, and local multiplicity

The one-variable results fit into ordinary algebraic geometry over K . We make the legitimate transfer explicit, because the proper-class notation **SC** by itself is not a substitute for a set-sized ring in a scheme construction; recall [remark 2.2](#).

15.1 The finite-type field lemma

Lemma 15.1 (Zariski's lemma in the needed form). *If an extension field L of an infinite field k is finitely generated as a k -algebra, then it is finite algebraic over k . In particular, if k is algebraically closed then $L = k$.*

Proof. Choose a maximal algebraically independent subcollection $t_1, \dots, t_r$ among a finite algebra generating set; the other finitely many generators are algebraic over $k(t_1, \dots, t_r)$, so L is a finite field extension of that rational-function field. Clearing the denominators of the coefficients in their monic algebraic equations supplies a nonzero $f \in k[t_1, \dots, t_r]$ such that every remaining generator is integral over $A = k[t_1, \dots, t_r, 1/f]$. Because L is generated as an algebra by the original generators and is a field containing $1/f$, it equals the algebra obtained by adjoining those integral generators to A ; thus L is integral over A .

If a field is integral over a subring A , that subring is a field: for $0 \neq a \in A$, a monic equation for $a^{-1} \in L$, multiplied by a suitable power of a , expresses a^{-1} as an element of A . If $r \geq 1$, choose $c \in k$ such that $t_1 - c$ does not divide f ; only finitely many linear factors divide a fixed nonzero polynomial, and k is infinite. Then $1/(t_1 - c) \notin A$, as one checks by multiplying a putative representation by a power of f and using divisibility in $k[t_1, \dots, t_r]$. This contradicts that A is a field. Hence $r = 0$, and the original generators are algebraic. $\square$

15.2 A class-safe Nullstellensatz for finite data

Theorem 15.2 (Surcomplex Nullstellensatz). *Let K be a set-sized algebraically closed Hahn workspace. A proper ideal $I \subset K[x_1, \dots, x_d]$ has a common zero in K^d , and $H \in K[x]$ vanishes at all common zeros of I if and only if $H^N \in I$ for some positive integer N .*

In class-safe form for finite data: let $F_1, \dots, F_s, G \in \mathbf{SC}[x_1, \dots, x_d]$ be finitely many finite-degree polynomials. Then G vanishes at every common surcomplex zero of the F_i if and only if, for some ordinary integer $N \geq 1$ and finite polynomials $H_i \in \mathbf{SC}[x_1, \dots, x_d]$,

$$G^N = \sum_{i=1}^s H_i F_i. \quad (15.1)$$

In particular the F_i have no common surcomplex zero if and only if 1 belongs to their polynomial ideal, and the certificate may be chosen inside any algebraically closed Hahn workspace containing all the coefficients.

Proof. Fix such a workspace K . A maximal ideal $\mathfrak{n}$ of the set ring $K[x]$ has field quotient finitely generated as a K -algebra, hence equal to K by [lemma 15.1](#); the images of the variables give coordinates $a_1, \dots, a_d \in K$ and $\mathfrak{n} = (x_1 - a_1, \dots, x_d - a_d)$. A proper ideal lies in such a maximal ideal and so has a common zero. Conversely, evaluation at a common zero is a homomorphism annihilating I , so $1 \notin I$.

If H vanishes at all common zeros, the ideal $J = IK[x, y] + (1 - yH)$ has no common zero, so the weak form gives $1 \in J$. Membership in this generated ideal uses only finitely many $F_i \in I$ and therefore gives an identity $1 = \sum_i A_i(x, y)F_i(x) + B(x, y)(1 - yH(x))$; no finite generating list for I has been assumed. If $H = 0$ the certificate is immediate; otherwise substitute $y = H^{-1}$ in the localization and multiply by a sufficiently high power H^N to clear denominators. This is the usual Rabinowitsch argument, the underlying classical theorem being recorded for instance in [[25](#), Tag 00FV].

For the class-safe form, the hypothesis about surcomplex zeros implies the hypothesis about K -zeros, so it supplies the certificate in K ; conversely that same finite identity persists in every larger workspace and hence at every surcomplex tuple. No proper-class maximal-ideal argument is used. $\square$

No priority claim attaches to this classical theorem. The point is that the field and ideal to which it is applied are specified and set-sized. A positive-dimensional solution set can acquire new points in a larger workspace, even though solvability does not change. The statement says nothing about an effective coefficient oracle for an arbitrary surreal constant: once coefficients belong to a computably presented field, ordinary finite algebraic algorithms may be available, but an arbitrary Hahn support need not admit such a presentation.

Remark 15.3 (A non-Noetherian witness). We have not asserted Noetherianity of any of the analytic coefficient rings in the supplied manuscripts, and already over a point it fails. For nonzero divisible Γ , the valuation ring $\mathcal{O}_K$ has the non-finitely generated ideal $\mathfrak{m}_K$: a finite set of positive generator valuations has a least member $\delta > 0$, but $t^{\delta/2}$ is not in the ideal they generate over $\mathcal{O}_K$. The positive part is not an ideal of the full Hahn field, where positive monomials are units. This distinction is essential, and it is exactly why [theorems 14.1](#) and [16.1](#) are proved over an arbitrary commutative ring with no finiteness hypothesis.

Proposition 15.4 (Finite fibers and local lengths). *If $B = K[x_1, \dots, x_d]/I$ is finite-dimensional of dimension $N > 0$, then its common-zero set $Z \subset K^d$ is finite and*

$$B \cong \prod_{\xi \in Z} B_\xi, \quad \sum_{\xi \in Z} \dim_K B_\xi = N, \quad (15.2)$$

each B_ξ being local with residue field K and nilpotent maximal ideal. Thus $|Z| \leq N$, with equality precisely when all local factors have dimension one. After extension to a larger algebraically closed workspace there are no new point locations and the local dimensions are unchanged. Moreover

$$\mathrm{Tr}(m_H) = \sum_{\xi \in Z} (\dim_K B_\xi) H(\xi), \quad \det(m_H) = \prod_{\xi \in Z} H(\xi)^{\dim_K B_\xi}. \quad (15.3)$$

Proof. Distinct maximal ideals are comaximal, so by the finite Chinese remainder theorem any s of them give a surjection $B \rightarrow K^s$, using [lemma 15.1](#) for their residue fields; hence there are at most N of them, and they correspond to Z by [theorem 15.2](#). Let J be their intersection. The chain J^m stabilizes by finite dimensionality; for a stable $M = J^m = JM$, choose finitely many generators, write each as a J -combination of them, and multiply the resulting matrix equation by its adjugate:

the determinant is $1 + j$ with $j \in J$, hence a unit annihilating all generators, so $M = 0$. The powers $\mathfrak{n}_\xi^m$ are pairwise comaximal with intersection $J^m = 0$, and the Chinese remainder theorem gives (15.2); each factor has one maximal ideal, nilpotent, with residue field K . After scalar extension each factor still has a nilpotent ideal and the extension field as quotient, so it is still local of the same dimension, and the coordinates modulo that ideal are the original tuple ξ . Finally, multiplication by $H - H(\xi)$ on each local factor is nilpotent, which gives (15.3). $\square$

The integer $\dim_K B_\xi$ is the algebraic local multiplicity, a zero-dimensional intersection multiplicity. For an arbitrary analytic system, constructing such a finite algebra is additional work and is not supplied by this proposition. A monic Gröbner basis computed over K remains one after extension to another field, since its division and critical-pair identities persist, so its standard monomials and quotient dimensions are unchanged; this is how elimination may be carried out in a fixed workspace, and it claims no algorithm for arbitrary surreal coefficients lacking finite descriptions.

Example 15.5 (A coupled infinitesimal four-point fiber). Let $t = \omega^{-1}$ and $f_1 = x^2 - ty - t^2$, $f_2 = y^2 - tx$. Since t is a nonzero field element, set $Y = y/t$; the second equation gives $x = tY^2$ and the first becomes $t^2(Y^4 - Y - 1) = 0$, so

$$K[x, y]/(f_1, f_2) \cong K[Y]/(Y^4 - Y - 1), \quad (15.4)$$

of dimension four, with $(1, x, y, xy) \mapsto (1, tY^2, tY, t^2Y^3)$. The ordinary polynomial $Y^4 - Y - 1$ has four distinct roots: a common root with $4Y^3 - 1$ would satisfy $Y^3 = 1/4$ and $Y/4 - Y - 1 = 0$, hence $Y = -4/3$, whose cube is not $1/4$. Its roots $c_1, \dots, c_4 \in \mathbb{C}$ are nonzero, so the actual solutions are $(x, y) = (tc_j^2, tc_j)$, all infinitesimal and simple, the Jacobian $4xy - t^2 = t^2(4c_j^3 - 1)$ being nonzero at each. In the stated basis

$$W_x = \begin{pmatrix} 0 & t^2 & 0 & 0 \\ 1 & 0 & 0 & t^2 \\ 0 & t & 0 & t^2 \\ 0 & 0 & 1 & 0 \end{pmatrix}, \quad W_y = \begin{pmatrix} 0 & 0 & 0 & t^3 \\ 0 & 0 & t & 0 \\ 1 & 0 & 0 & t^2 \\ 0 & 1 & 0 & 0 \end{pmatrix},$$

which commute and satisfy $W_x^2 = tW_y + t^2I$, $W_y^2 = tW_x$; eliminating x gives the resultant $y^4 - t^3y - t^4$. A coupled system whose ordinary reduction is $x^2 = y^2 = 0$ thus splits into four actual surcomplex points while its finite-algebra length stays four.

16 Degree-controlled complete intersections over arbitrary rings

Only one of the three merged manuscripts leaves one variable, and this section and the next are its contribution. Let A be a nonzero commutative ring with identity and

$$F_i(x) = x_i^{d_i} + E_i(x) \in A[x_1, \dots, x_n], \quad d_i \geq 1, \quad \deg E_i < d_i \quad (1 \leq i \leq n), \quad (16.1)$$

where throughout this section and the next, degree means *total* degree. Put

$$\mathcal{B} = \{\alpha \in \mathbb{N}^n : 0 \leq \alpha_i < d_i\}, \quad D = \prod_i d_i, \quad \rho_\# = \sum_i (d_i - 1), \quad x^{\text{top}} = \prod_i x_i^{d_i-1}.$$

The errors may be fully coupled and their coefficients may have arbitrary size or valuation: the hypothesis is on polynomial degree, not on smallness. The degree hypothesis is a genuine restriction and not a disguised assertion about all isolated complete intersections.

Theorem 16.1 (Global finite-free normal forms). *For (16.1), the quotient $B = A[x]/(F_1, \dots, F_n)$ is free over A with basis $(x^\alpha)_{\alpha \in \mathcal{B}}$ and rank D . Every polynomial H has a unique normal form in their span and a finite division identity*

$$H = \sum_i F_i Q_i + \text{NF}_F(H). \quad (16.2)$$

Reduction never increases total degree, and every actual use of a relation strictly lowers the degree of the replaced monomial. All output coefficients are polynomial expressions with integer coefficients in the input coefficients, so all of this commutes with every ring homomorphism on coefficients. The sequence $F_1, \dots, F_n$ is regular in any order.

Proof. Use the rules $x_i^{d_i} \mapsto -E_i$; every replacement has smaller total degree, so all branches of reduction terminate after finitely many steps on a monomial, and by linearity on every polynomial. The monomials on which no rule applies are exactly x^α , $\alpha \in \mathcal{B}$.

For compatibility, without assuming coefficients form a field: an ambiguity involving the rules for $i \neq j$ occurs on a multiple of $x_i^{d_i} x_j^{d_j}$; reducing the two factors in one order gives a multiple of $E_i E_j$ and in the other a multiple of $E_j E_i$, the same polynomial, and any remaining reductions take place in smaller total degrees. Induction on total degree resolves these ambiguities, and disjoint monomial reductions commute by linearity. Equivalently these are monic leading terms with relatively prime leading monomials whose critical-pair differences reduce to zero. No division by a coefficient occurs, so the completely reduced polynomial is independent of the reduction choices.

Each rule subtracts a multiple of one F_i , giving (16.2); conversely $x_i^{d_i} Q$ and $-E_i Q$ have the same normal form, so the normal-form map annihilates $F_i Q$ and hence the ideal, and an already reduced polynomial in the ideal is zero. This proves both spanning and independence. The coefficients are obtained by finitely many additions, subtractions and multiplications in A , hence commute with all coefficient homomorphisms.

For regularity, apply the same reduction to any initial subset of the equations; its quotient has standard monomials with the corresponding exponents bounded and the remaining exponents unrestricted, and because reduction strictly lowers degree its total-degree associated graded ring is $A[x]/(x_i^{d_i} : i \text{ in that subset})$. Multiplication by the highest part $x_j^{d_j}$ of the next equation is injective there, merely shifting the still unrestricted exponent of x_j ; so the highest nonzero graded term of a nonzero element survives multiplication by F_j , and F_j is a non-zero-divisor in the preceding quotient. The final quotient is nonzero, being free of positive rank over nonzero A . The argument works in every ordering. $\square$

Corollary 16.2 (A global Bézout count, including infinite-scale roots). *Over an algebraically closed Hahn field K , the system (16.1) has total local algebraic multiplicity D in all of $\mathbf{SC}^n$. If its Jacobian determinant is nonzero at every solution, there are exactly D distinct solutions. The homogenized system has no additional projective solutions at infinity.*

Proof. Use the rank- D quotient and (15.2); workspace independence (proposition 2.1) includes every surcomplex solution. At a solution with invertible Jacobian the images of the equations generate the cotangent space of the coordinate maximal ideal, so Nakayama's lemma in the finite local algebra makes its maximal ideal zero and that factor has dimension one. At the projective hyperplane $x_0 = 0$ the homogenized equations read $x_i^{d_i} = 0$ for every i , forcing all projective coordinates to vanish, which is impossible. $\square$

An infinite surcomplex coordinate is still an affine coordinate in this statement; it is not the projective point at infinity, and the last sentence does not assert that all coordinates have nonnegative valuation.

16.1 Finite support propagation without positivity

Let A be a Hahn coefficient ring, S contain the coefficient support of H , and T be the union of the coefficient supports of the E_i . Neither support need be positive. With $L = \deg H \geq 0$,

$$\text{supp NF}_F(H) \cup \bigcup_i \text{supp } Q_i \subseteq S + \bigcup_{k=0}^L kT, \quad 0T = \{0\}, \quad (16.3)$$

the empty polynomial having zero outputs. Indeed, following one monomial through the reduction, each use of an error adds one input support letter and decreases total degree by at least one, so at most L such uses occur on a branch; the witness polynomials Q_i accumulate only the preceding multipliers and obey the same bound. Finite unions and finite Minkowski sums of well-ordered sets are well ordered, and the coefficientwise products have finite multiplicity by [lemma 2.4](#).

This finite-degree mechanism differs from the infinite positive-monoid inverse S^* of [theorem 12.1](#). It permits *negative* valuations without sacrificing support control, but only because total degree rules out an infinite reduction chain. Neither mechanism subsumes the other: analytic perturbations need not have bounded degree, while polynomial perturbations need not have positive support.

17 Multivariate Bézoutians and the Jacobian trace formula

Retain [\(16.1\)](#) over the arbitrary commutative ring A and define

$$\lambda_F(H) = [x^{\text{top}}] \text{NF}_F(H). \quad (17.1)$$

Bézoutian and residue duality are classical subjects [\[23, 24\]](#); what is proved here is an explicit statement for this family over arbitrary coefficient rings, with the comparison to the supplied Hahn-coherent framework.

Theorem 17.1 (A coefficient-independent perfect pairing). *The pairing $(G, H) \mapsto \lambda_F(GH)$ on B is perfect over A . Its Gram determinant in the rectangular basis is the sign of the permutation $\alpha \mapsto \mathbf{d} - \mathbf{1} - \alpha$, hence a constant $+1$ or -1 independent of every coefficient of the errors. For $\deg H < \rho_\#$ one has $\lambda_F(H) = 0$, whereas $\lambda_F(x^{\text{top}}) = 1$.*

Proof. Reduction cannot increase degree, so $\lambda_F(x^\alpha x^\beta) = 0$ when $|\alpha| + |\beta| < \rho_\#$. If the degrees sum to $\rho_\#$, reduction of any nonstandard monomial strictly lowers degree, so

$$\lambda_F(x^{\alpha+\beta}) = \begin{cases} 1, & \alpha + \beta = \mathbf{d} - \mathbf{1}, \\ 0, & \text{otherwise,} \end{cases} \quad |\alpha| + |\beta| = \rho_\#. \quad (17.2)$$

Order the row indices α by nondecreasing total degree and the columns by the complementary indices $\mathbf{d} - \mathbf{1} - \alpha$. The Gram matrix is then block lower triangular with identity diagonal blocks: degrees below the row's complementary level give zero, equal degrees give [\(17.2\)](#). Its determinant in this reordering is one; undoing the complement permutation gives the stated determinant and therefore perfection over A . The last two assertions are the same degree argument. $\square$

17.1 The divided-difference determinant

For two coordinate tuples X, Y define

$$D_{ij}(X, Y) = \frac{F_i(Y_1, \dots, Y_{j-1}, X_j, \dots, X_n) - F_i(Y_1, \dots, Y_j, X_{j+1}, \dots, X_n)}{X_j - Y_j}, \quad (17.3)$$

each numerator being divisible by its displayed denominator as a polynomial identity over A , and put

$$\Delta_F(X, Y) = \det(D_{ij}(X, Y)), \quad J_F(x) = \det\left(\frac{\partial F_i}{\partial x_j}(x)\right), \quad (17.4)$$

viewing Δ_F in $B \otimes_A B = A[X, Y]/(F(X), F(Y))$.

Theorem 17.2 (Bézoutian kernel and Jacobian duality). *Let $(b_\alpha) = (x^\alpha)_{\alpha \in \mathcal{B}}$ and let (b^α) be its residue-dual basis for λ_F . Then*

$$\Delta_F(X, Y) = \sum_{\alpha \in \mathcal{B}} b_\alpha(X) b^\alpha(Y) \quad \text{in } B \otimes_A B, \quad (17.5)$$

$$J_F = \sum_{\alpha \in \mathcal{B}} b_\alpha b^\alpha \quad \text{in } B, \quad (17.6)$$

$$\text{Tr}_B(m_H) = \lambda_F(J_F H) \quad (H \in B). \quad (17.7)$$

These are identities over A , including nonreduced coefficient rings and parameters at which roots collide.

Proof. The columnwise telescoping identity is $D(X, Y)(X - Y) = F(X) - F(Y)$; multiplying by the adjugate of D and passing to $B \otimes_A B$ gives

$$(X_j - Y_j)\Delta_F = 0 \quad (1 \leq j \leq n), \quad (17.8)$$

with no matrix inverse or determinant denominator anywhere in the argument.

The total degree of row i of D is at most $d_i - 1$, and the highest-degree contribution of that row is diagonal, coming from $x_i^{d_i}$. So Δ_F has total degree at most $\rho_\#$, with homogeneous part of degree $\rho_\#$ equal to

$$\prod_{i=1}^n \left(\sum_{k=0}^{d_i-1} X_i^{d_i-1-k} Y_i^k \right). \quad (17.9)$$

Apply λ_F in the X variables: terms of X -degree below $\rho_\#$ vanish by [theorem 17.1](#), and a nonstandard X -monomial of degree $\rho_\#$ reduces to smaller degree and also vanishes. The remaining monomial is X^{top} , whose coefficient in [\(17.9\)](#) is one with no Y dependence, so

$$(\lambda_F \otimes \text{id})(\Delta_F) = 1. \quad (17.10)$$

By [theorem 17.1](#) the kernel map [\(14.9\)](#), with B and λ_F in place of A_P and λ_P , is an isomorphism $B \otimes_A B \cong \text{End}_A(B)$. Equation [\(17.8\)](#) says the operator attached to Δ_F commutes with multiplication by each coordinate, hence, the coordinates generating B as an A -algebra, that it is B -linear; [\(17.10\)](#) says it sends 1 to 1. It is therefore the identity, whose unique representing tensor is the right side of [\(17.5\)](#). The multiplication map $B \otimes_A B \rightarrow B$ substitutes $X = Y = x$, turning each divided difference into the corresponding partial derivative, so the diagonal of Δ_F is J_F , which is [\(17.6\)](#). Finally the coefficient of b_α in Hb_α is $\lambda_F(Hb_\alpha b^\alpha)$; summing these diagonal matrix coefficients gives [\(17.7\)](#). $\square$

Corollary 17.3 (Simple-root weights and Euler–Jacobi vanishing). *Over an algebraically closed Hahn field K , suppose every solution of (16.1) is simple. Then*

$$\lambda_F(H) = \sum_{\xi \in Z} \frac{H(\xi)}{J_F(\xi)}, \quad (17.11)$$

and consequently

$$\sum_{\xi \in Z} \frac{H(\xi)}{J_F(\xi)} = 0 \quad \text{whenever } \deg H < \rho_{\sharp}, \quad \sum_{\xi \in Z} \frac{\xi^{\text{top}}}{J_F(\xi)} = 1. \quad (17.12)$$

For arbitrary multiplicities, with e_{ξ} the local-factor idempotent of (15.2), the functional $\lambda_{F,\xi}(H) := \lambda_F(e_{\xi}H)$ is a perfect local residue functional and

$$\lambda_{F,\xi}(J_F H) = (\dim_K B_{\xi}) H(\xi). \quad (17.13)$$

Proof. Different local factors are orthogonal for the residue pairing, their products being zero, so its restriction to each factor is perfect. For a simple factor Ke_{ξ} write $\lambda_F(e_{\xi}H) = c_{\xi}H(\xi)$; the trace identity (17.7) applied to e_{ξ} gives $1 = c_{\xi}J_F(\xi)$, which is (17.11). The degree bounds of theorem 17.1 then give (17.12). In a factor of length μ multiplication by H has trace $\mu H(\xi)$, and the global identity applied to $e_{\xi}H$ gives (17.13). $\square$

The simple-root formula is not used to define the functional. Defining it that way would obscure the multiple-root case and would falsely suggest that division by a discriminant is necessary. The top-normal-form definition (17.1), the perfect Gram matrix, and the Bézoutian all remain defined at every collision — exactly as in one variable, where theorem 14.1 has no discriminant denominator either.

17.2 Parameters on one fixed ordinary domain, and the analytic comparison

Theorem 17.4 (Global polynomial families on one ordinary parameter domain). *Let $U \subseteq \mathbb{C}^s$ be one fixed ordinary domain and suppose the finitely many coefficients in (16.1) lie in $A = \mathcal{H}_{\Gamma}(U)$ or in $A = \mathcal{H}_{\Gamma, \geq 0}(U)$. Then the global polynomial quotient is finite free of rank D over A ; normal forms, multiplication matrices, the residue Gram matrix and its inverse, and the Bézoutian trace identity all have coefficients holomorphic on the same U . They commute with every allowed finite coherent parameter substitution and with specialization to a finite surcomplex parameter. No positivity of the errors is required for the version over $\mathcal{H}_{\Gamma}(U)$, and no ordinary neighbourhood is shrunk.*

Proof. Apply theorems 16.1, 17.1 and 17.2 to the indicated coefficient ring. Normal-form computations are finite, obey (16.3), and only add and multiply functions already holomorphic on U . The inverse Gram matrix is its adjugate divided by the constant determinant ± 1 , so it too involves only finite polynomial operations, and all other constructions are finite determinants or matrix operations. Since evaluation and allowed coherent parameter substitutions are ring homomorphisms, the base-change assertion of theorem 16.1 gives compatibility. $\square$

Now suppose in addition that every coefficient of every E_i has *positive* Hahn support, so that the parameter ring is integral. On an ordinary product of circles in the fiber variables, the supplied research manuscript defines its residue by the strong coefficient formula

$$\Lambda_F(H) = \sum_{k \in \mathbb{N}^n} (-1)^{|k|} \left[x_1^{d_1(k_1+1)-1} \cdots x_n^{d_n(k_n+1)-1} \right] \left(H \prod_i E_i^{k_i} \right), \quad (17.14)$$

obtained by expanding $1/F_i$ about $x_i^{d_i}$ and applying the ordinary Cauchy coefficient formula at each Hahn exponent; positive supports make this family strongly summable by [lemma 2.4](#). Its values are coherent parameter functions and $\Lambda_F(F_i H) = 0$, since after cancellation of F_i the coefficientwise integrand is holomorphic in x_i on its disk. These are the functional and annihilation property proved in [\[5\]](#).

Theorem 17.5 (Polynomial identification of the Jacobian trace element). *Under the lower-total-degree hypothesis of [\(16.1\)](#) and the positive-support hypothesis just stated — positivity being required of the errors E_i , not of the ambient ring — over the fixed-domain ring $\mathcal{H}_{\Gamma, \geq 0}(U)$,*

$$\Lambda_F(H) = \lambda_F(H) \quad \text{for every polynomial } H. \quad (17.15)$$

Consequently the trace element of the supplied residue pairing, restricted to this polynomial family, is exactly the class of J_F , so that $\text{Tr}(m_H) = \Lambda_F(J_F H)$. All solutions in a specialized surcomplex fiber are infinitesimal, so the global polynomial count D agrees with the finite-monad analytic count in this subclass.

Proof. Evaluate [\(17.14\)](#) on a standard monomial $H = x^\alpha$, of degree at most $\rho_\sharp$. For $k \neq 0$ the extraction degree is $\rho_\sharp + \sum_i d_i k_i$, whereas the numerator has degree at most $|\alpha| + \sum_i (d_i - 1)k_i \leq \rho_\sharp + \sum_i d_i k_i - |k|$, strictly smaller. So every $k \neq 0$ term vanishes, and the $k = 0$ term is one precisely for the top monomial and zero for the other basis monomials. Since both functionals annihilate the defining ideal, normal forms extend the equality to every polynomial; their Gram matrices in the shared rectangular basis coincide, so the element representing the trace under the perfect pairing is J_F by [theorem 17.2](#).

For the location of the roots, specialize the parameters in an algebraically closed Hahn workspace. The multiplication matrices of the x_i have integral coefficients, and reducing their entries modulo $\mathfrak{m}_K$ gives the nilpotent coordinate shifts in $K[x]/(x_1^{d_1}, \dots, x_n^{d_n})$ reduced over $\mathbb{C}$. Each characteristic polynomial therefore has the form $T^D + \sum_{j < D} c_j T^j$ with every $c_j \in \mathfrak{m}_K$. A noninfinitesimal root would have finite inverse, and dividing by its D th power would express zero as one plus a finite sum of infinitesimals, which is impossible. All eigenvalues, hence all solution coordinates, are infinitesimal. The finite polynomial quotient and the analytic quotient have the same rectangular basis and the same defining multiplication relations, so the inclusion of polynomials induces an isomorphism between them. $\square$

This settles a precisely delimited comparison left unproved in the supplied research manuscript: the Jacobian identity for its residue functional in the lower-degree polynomial family, over the fixed-domain ring. It does *not* settle the comparison for arbitrary holomorphic errors, nor for polynomials with higher-degree positive errors that can create roots outside the finite monad — [example 2.8](#) is a warning against exactly that extrapolation. The general classical theory of complete-intersection residues already contains Jacobian identities; the contribution here is a direct normalization and support-compatible identification with the particular supplied functional, not a claim that the classical identity was previously unknown.

Example 17.6 (A six-dimensional coupled algebra with an explicit dual basis). Let a, b, c, d be arbitrary surcomplex coefficients, or indeterminates in a commutative coefficient ring, and take

$$F_1 = x^2 - ay - b, \quad F_2 = y^3 - cx - d, \quad (17.16)$$

so $(d_1, d_2) = (2, 3)$ and the degree hypotheses hold. The quotient has rank six with basis $(1, y, y^2, x, xy, xy^2)$ and $\lambda_F(H) = [xy^2] \text{NF}_F(H)$; no nonzero condition on a, b, c, d is required. For a

second tuple (u, w) ,

$$D(X, Y) = \begin{pmatrix} x+u & -a \\ -c & y^2 + yw + w^2 \end{pmatrix}, \quad \Delta_F = (x+u)(y^2 + yw + w^2) - ac,$$

whose coefficients in the first basis give the residue-dual basis

$$(xy^2 - ac, xy, x, y^2, y, 1), \quad (17.17)$$

with diagonal $J_F = 6xy^2 - ac$ and residue Gram determinant -1 for every coefficient choice. Thus $\text{Tr}(m_H) = \lambda_F((6xy^2 - ac)H)$, including at all collisions. When $c \neq 0$ ordinary elimination gives $y^6 - 2dy^3 - ac^2y + d^2 - bc^2 = 0$ and $x = (y^3 - d)/c$; at $c = 0$ it is the division by c that fails, not the six-element basis or the residue theory — a global basis avoids exceptional denominators introduced by a particular elimination.

For a concrete infinitesimal splitting take $a = c = t > 0$ and $b = d = 0$. Solving $x = y^3/t$ gives $y(y^5 - t^3) = 0$, so the roots are $(0, 0)$ and $(t^{4/5}\zeta^3, t^{3/5}\zeta)$ with $\zeta^5 = 1$. The Jacobian is $-t^2$ at the origin and $5t^2$ at each of the other five, all six roots being simple, and

$$\lambda_F(H) = -\frac{H(0,0)}{t^2} + \frac{1}{5t^2} \sum_{\zeta^5=1} H(t^{4/5}\zeta^3, t^{3/5}\zeta). \quad (17.18)$$

Each individual weight is infinite, yet $\lambda_F(1) = 0$ by cancellation and $\lambda_F(xy^2) = 1$. In the underlying parameter family the value $a = c = b = d = 0$ gives the length-six origin with algebra $K[x, y]/(x^2, y^3)$ and the same nondegenerate residue determinant.

18 Polynomials versus Hahn-coherent entire data

There is a simple exact description of the intersection of the two languages. For fixed Γ ,

$$K_\Gamma[z] = \bigcup_{d \geq 0} (\mathbb{C}[z]_{\leq d}((t^\Gamma))), \quad (18.1)$$

equality of polynomial coefficient data, or of the corresponding Hahn-coherent evaluations on the finite thickened plane. A finite-degree polynomial has finitely many Hahn coefficients whose finite union of supports is well ordered, and grouping by exponent gives ordinary polynomial coefficient functions with the same uniform degree bound; conversely a Hahn family of ordinary polynomials of degree at most d regroups into $d + 1$ Hahn coefficients whose supports are subsets of the given common well-ordered support.

The bound must be uniform in the exponent. The coherent datum (2.5) has ordinary entire coefficient functions and well-ordered support, yet no uniform degree bound; on finite surcomplex points it equals $1/(1 - tz)$ by strong geometric summation, and it is not a polynomial, since its ordinary coefficient functions have unbounded degrees and coefficient uniqueness excludes a bounded-degree representation. Its rational continuation has a pole at the infinite point t^{-1} . Thus “entire on the thickened ordinary plane” is not “entire at all surreal scales”; this is consistent with the all-scale polynomial rigidity theorem stated in [4], which we take as given and do not reprove.

For a finite root multiset, the supports of its finitely many symmetric coefficients have well-ordered union, so the finite CRT, cluster-factor and residue constructions above never encounter a failure caused merely by an infinite number of spatial centers. They can still produce negative

exponents through small denominators, and root extraction can enlarge the exponent group. The supplied global manuscript [6] proves a different phenomenon for infinitely many ordinary centers: its cluster family $P_m(u) = u - t^{1/m}$ cannot be the exact zero divisor of one Hahn-coherent entire datum, because the union of the symmetric coefficient supports is decreasing, whereas its family $P_m(u) = u^m - t$ is realizable, all symmetric coefficients using a single exponent even though the individual root supports contain the descending sequence $1/m$. These are results of that supplied manuscript, not newly established global theorems here. A claim that every separately specified infinite family can be assembled from the finite results above would be unjustified.

Finally, three mechanisms have been used and are *not* interchangeable. [Theorem 6.1](#) quantifies over an order circle and uses a fixed-degree first-order statement. [Corollary 8.3](#) compares finite coefficient valuations and preserves an initial polynomial. The supplied contour theory integrates ordinary holomorphic coefficients and then assembles a strongly summable Hahn series. Agreement of some consequences does not make the mechanisms interchangeable.

19 Conflicts recorded from the wider set

The merge contract governing this report records two disagreements in the wider manuscript set that touch this cluster. Their mathematics belongs to the companion report on residues and contours, and is not restated here; what is recorded is their *status*, because the present report's residue functionals are the polynomial restrictions of the objects at issue and it would be easy to read more into them than is proved.

19.1 Whether an arbitrary isolated complete intersection has a contour representation of its perturbation residue

Three manuscripts of the wider set treat this differently, and the disagreement is about status, not arithmetic: the same statement is proved, assumed, and declined.

One proves it unconditionally. With A constructed from finite colength alone so that $AF = (z_1^{e_1}, \dots, z_n^{e_n})$ and $G = AF$, it establishes $\text{Res}_F(H) = \text{Res}_G(H \det A)$ and then a Hahn integral representation of $\text{Res}_F(H)$ over the coordinate torus T_r as $(2\pi i)^{-n} \int_{T_r} H \det A / (G_1 \cdots G_n)$, needing only the strong residue expansion, ordinary finite-family residue transformation and positive rescaling. Its own two non-claims survive as well: the *untransformed* form has no such torus integral, and $\det A$ is not an optional Jacobian factor, since AF can introduce extra zeros.

A second reaches a determinant torus formula too, but its theorem opens by assuming the finite-algebra and normal-form results of a companion geometry manuscript and routes through that companion's univariate separating denominators. Its general-case result is therefore *conditional* on an unrefereed companion, not independent.

A third explicitly *declines* the same statement, proving the realization for one variable and for a coordinate-power class only, and arguing that for arbitrary isolated f a single diagonal rescaling need not make the leading equations coordinate powers or even keep the leading homogeneous system isolated; that the usual cycle $\{|f_i| = \epsilon_i\}$ is not a coordinate product torus; and that only a germwise, coefficient-dependent cycle interpretation is available. It lists a general realization theorem as an open problem.

Resolution as applied here. We do not average these. The proved theorem and its proof belong in the companion contours report, printed there with the third manuscript's scope discussion reproduced immediately afterwards, stating plainly that one of the three sources judged the general

case unproved and why, and that the second’s route is conditional rather than independent. For the present report the consequence is a scope statement: *nothing in this article asserts a contour representation of any residue*. The functional λ_P of (14.1) and the functional λ_F of (17.1) are defined by finite polynomial division, and [proposition 14.4](#) and [theorem 17.2](#) are proved from that definition alone. The one place where a contour functional appears is [theorem 17.5](#), which identifies Λ_F with λ_F only under both the lower-total-degree hypothesis and the positive-support hypothesis, over the fixed-domain ring, and which says explicitly that the general analytic comparison needs another argument.

19.2 Whether roots-of-unity quadrature recovers the Hahn contour integral

Two manuscripts of the wider set leave opposite impressions here, and the hypothesis is doing all the work.

One treats it as a negative result. Its example takes $f(z) = 1/(z - b)$ with ordinary real $b = 1/2$ on the unit circle, computes the N -point roots-of-unity quadrature $Q_N(f) = 1/(1 - b^N)$, and observes that although the ordinary complex limit is 1, one has $v(Q_N(f) - 1) = 0$ for every N : there is no convergence in the Hahn valuation topology, and none in the full fine topology either, since the family is not eventually constant. With $b = t^\varrho$ a rank-one valuation can see convergence, but a higher-rank group with an element exceeding every $N\varrho$ still cannot.

The other devotes a section to the positive side, with a rank-independent aliasing criterion and a coefficientwise recovery theorem, under the decay condition that for every γ the set of k with $v(A_k) \leq \gamma$ is finite.

Resolution as applied here. The two are compatible; the first even concedes in prose that taking ordinary coefficient limits gives the Hahn functional, and that this is a legitimate alternative description but not a limit in the internal valuation topology. The positive theorem is to be stated *with* its decay hypothesis, and the negative example kept immediately alongside as the demonstration that the hypothesis cannot be dropped: the decay condition demands eventual domination of *every* value-group threshold, and, as its own source notes, cannot be met by a nontrivial sequence of finite valuations when no countable subset is cofinal in the positive value group. For the present report the consequence is again a scope statement, and one this article has enforced from [remark 2.5](#) onward: no construction here is justified by a quadrature, a limit of partial sums, or an argument that a valuation improves by a fixed positive amount at each iteration.

20 Scope, provenance, and what is not proved

20.1 A consolidated status statement

All three merged manuscripts carry a status disclaimer of the same shape. They are consolidated here into one statement, which governs the whole article.

This is a theorem-driven research exposition. It supplies proofs of the stated analogues and support estimates. It does not claim that these are first occurrences in the literature, does not certify historical priority, and does not claim to resolve a named published conjecture. Several central statements are classical over real-closed fields, valued fields, or arbitrary commutative rings, and are identified as such. The literature search behind these attributions was conducted on 21 September 2026 and was targeted rather than exhaustive. The article has not been independently refereed. The original manuscript packages were not proof-assistant formalizations; the repository’s separate coverage ledger now records partial Lean results, without asserting

a complete formalization of this article. The symbolic checks recorded in the source archives use exact rational, polynomial and finitely truncated formal-series inputs to validate *displayed finite examples and identities*; they are not machine verification of the general proofs, and in particular they certify nothing about arbitrary Hahn supports, arbitrary-rank exponent groups, transfinite summability, class-set foundations, or real-closed-field quantifier elimination. The supplied predecessor manuscripts [4, 5, 6] are treated as unpublished source material, credited for their Hahn-coherent framework, their analytic antecedents and their reused examples, and are not treated as independently verified publications. No theorem is declared new merely because the word “surcomplex” is placed in its title.

20.2 Per-theorem non-claims

The per-theorem non-claims of the three sources are specific, are *not* pooled into the statement above, and are preserved where they were made. They are listed here for reference.

- **Classical inputs.** Euclidean division, finite Chinese remainders, resultants, the Nullstellensatz and residue duality are classical algebra. Gauss–Lucas is classical complex polynomial geometry, and the proof of [theorem 4.2](#) works over any real closed field. Real-closed-field transfer and non-Archimedean root geometry have extensive precedents [11, 14, 21, 13]. It would be misleading to label their surcomplex instances wholly new. [Theorem 3.1](#) is an application of established algebraic closedness, not a new proof of the surreal normal-form theorem.
- **Support calculus.** [Lemma 2.4](#) does *not* say that the relevant partial sums converge in the fine surreal topology, and does *not* say that $k\rho$ eventually exceeds every positive value-group element; the latter is false in higher rank. Neither false cofinality claim is ever substituted for the lemma. See [remarks 2.5](#) and [2.6](#).
- **No proper-class objects.** No assertion treats a proper-class-indexed sum, an infinite-degree object as a polynomial, or a strong Hahn sum as the limit of its partial sums in the full fine topology. The formal expression (2.5) is not a global surcomplex polynomial; all global polynomial evaluations here are finite sums. See [remarks 2.2](#) and [2.7](#).
- **Gauss–Lucas.** The proof of [theorem 4.2](#) is finite and algebraic; it does not call on compactness of a convex hull or a topological separation theorem. Neither its conclusion nor that of [proposition 4.3](#) describes how critical points are distributed among infinitesimal clusters.
- **Jensen disks.** [Theorem 4.4](#) concerns F -valued disks; it is not a statement about ordinary pictures obtained by discarding infinitesimals.
- **Schoenberg.** [Theorem 5.3](#) is the surcomplex version of a classical inequality, not a newly discovered general principle; its matrix proof does not assert completeness of an infinite-dimensional surcomplex Hilbert space.
- **Transfer.** [Section 6.1](#) states a theorem *scheme*, one sentence for each fixed finite degree bound. It does not transfer unrestricted quantification over functions, sequences, arbitrary subsets, or the standard natural numbers as a distinguished definable set, and it does not make a proper class into a first-order set-sized model. [Theorem 6.1](#) is a transfer theorem, not a new analytic proof; it does not define an integral along a fine-continuous real-parameter surreal circle, and it is not the same statement as the leading-coefficient Rouché theorem on ordinary thickened domains in [4], whose hypothesis is different. No real-parameter fine-continuous circle or contour integral has been introduced anywhere in this article.

- **Sampling.** [Example 6.3](#): the exact quantifier in (6.1) cannot be replaced by sampling ordinary points.
- **Two Rouché theorems.** The modulus and valuation versions have different hypotheses and different balls. Neither is identified with the contour theorem on an ordinary thickening in [4].
- **Hermite signatures.** [Theorem 6.4](#) is a classical signature method, related to the residue and Bézoutian forms of [23]; it is *not* the nondegenerate residue pairing of [section 14](#).
- **Newton polygons.** In [theorem 9.1](#) and [corollary 8.3](#), sufficiency of the tropical double-minimum condition is a consequence of algebraic closedness together with the exact count, *not* an additional convergence assertion about a Newton algorithm.
- **Asymptotic notation.** In [examples 9.2](#) and [12.4](#), $O(t^k)$ means an integer-power formal Hahn series supported in exponents at least k , not a fine-topological limiting estimate and not a numerical asymptotic assertion obtained by varying t .
- **Nondiscrete supports.** In [example 9.4](#), the binomial identity does not require that the coefficient series converge after replacing t by a positive real number $q \in (0, 1)$; in fact $q^{\gamma_k} \rightarrow q$, not 0, so the summands would fail to tend to zero. No such substitution is used.
- **Differentiation.** The hypothesis of [lemma 11.1](#) cannot simply be removed: a constant initial polynomial has zero derivative and does not determine the next nonzero scale of P' . The argument can fail in positive residue characteristic, where a nonconstant reduction may have zero derivative, and [remark 11.3](#) records that a root-free ball is a genuinely different case.
- **Critical allocation.** [Corollary 11.7](#) is an allocation theorem, not a claim that pairwise root valuations alone determine every critical point; a residual critical polynomial with a multiple root may require its own rescaling analysis. [Corollary 11.5](#) is not a proof of a Euclidean root-critical-point distance conjecture.
- **Stability.** [Theorem 13.3](#) does not assert equality of the roots, of their ordinary moduli, or of every Hahn coefficient of the pairwise differences; [corollary 13.7](#) preserves the allocated critical directions, not the individual multiplicities of all critical points. [Example 13.5](#) proves sharpness of the strict boundary *in that family*; it does not claim that $v(\text{Disc } P)$ is the optimal threshold for every polynomial, and [example 13.6](#) shows it is not.
- **Residues.** [Theorem 14.1](#) is the polynomial core of the one-variable residue duality in [5]; it is not claimed as a new general duality theorem. [Theorem 14.2](#) is a statement about the monogenic algebra $R[z]/(P)$ and is not a proof of the general multivariable Jacobian comparison. For coherent analytic numerators of unbounded Taylor degree one must first use the fixed-domain division theorems of the supplied manuscripts; the polynomial argument of [section 14.4](#) does not replace that step.
- **Several variables.** [Theorem 17.5](#) does not settle the comparison for arbitrary holomorphic errors, nor for polynomials with higher-degree positive errors that can create roots outside the finite monad; [example 2.8](#) is the warning against that extrapolation. The degree hypothesis in (16.1) is a genuine restriction, not a disguised assertion about all isolated complete intersections.
- **Effectivity.** A finite algebraic operation on arbitrary Hahn inputs is well defined but not automatically computationally effective. A support may have no supplied computable enumeration, and equality of arbitrary surreal coefficients may have no finite decision procedure in the chosen representation. Noetherianity of the analytic coefficient rings of the supplied manuscripts is not asserted; see [remark 15.3](#).

- **Contours.** Residues here are either finite formal Laurent coefficients or, in [theorem 17.5](#) alone, the specifically identified coherent contour functional; those interpretations are never conflated with integration along fine-continuous real-parameter loops. See also [section 19](#).

20.3 Hypotheses that cannot be discarded

Degree is always finite; without that assumption neither the finite factorization nor the finite root partitions and determinant proofs are statements of the same kind. Divisibility of the exponent group is used wherever roots, midpoints of scales or slopes are needed, and no claim is made that $K_{\mathbb{Z}}$ is algebraically closed. Positive support in [theorem 12.1](#) is a *common well-ordered support* condition, not merely a pointwise assertion about an unrelated family of errors; coprimality there is essential for the support conclusion. The exact critical count requires a ball occupied by at least one root of the polynomial being counted, and the mapping-ball form uses $P - P(a)$ to ensure that occupation. All derivative arguments use residue characteristic zero, through (2.3). Integral coefficients in the perturbation estimates ensure that translating by a finite root preserves the coefficient error bound, and [corollary 10.2](#) and [section 13.3](#) record the change of coordinates that makes this harmless. The ordered Gauss–Lucas theorem uses a real closed ordered real part and its conjugation; algebraic closedness alone does not define its convex hull. The global complete-intersection theorems use strict lower total degree. No general closed-form solution by radicals is asserted anywhere, and the presence of surreal coefficients does not make ordinary higher-degree Galois obstructions disappear in a prescribed smaller coefficient field.

20.4 A consolidated theorem map

Classical principle	Surcomplex conclusion	Mechanism, ring, and status
Fundamental theorem of algebra	Finite factorization in a set-sized Hahn workspace	Established Hahn algebraic closedness, then finite induction, over K ; theorem 3.1 .
Gauss–Lucas, Jensen, weighted incomplete polynomials	Convex-hull, conjugate-disk and exact-hull statements with surreal weights	Finite reciprocal identities over F and K ; theorems 4.2 and 4.4 and proposition 4.3 .
Schoenberg inequality	Sharp second-moment bound with collinearity equality case	Finite matrix compression over a real closed field; theorem 5.3 . Classical inequality.
Polynomial Rouché; Hermite signatures	Order-circle counts at arbitrary surreal radii; real-root counts as signatures	Precisely encoded real-closed-field transfer, and finite trace forms; theorems 6.1 and 6.4 .
Pseudozero sets	<i>Exact</i> root loci under bounded coefficient error, in both geometries	Explicit phase construction and a single-coefficient witness, over SC and K ; theorem 7.1 .
Newton polygon	Counts in every ball, shell and residue direction, at arbitrary group rank	One finite initial-polynomial product over K ; theorems 8.2 and 9.1 .
Non-Archimedean disk mapping	Exact image balls and two mapping degrees	The same product theorem applied to $P - y$, over K ; theorem 10.1 .

Classical principle	Surcomplex conclusion	Mechanism, ring, and status
Finite flat maps and Riemann–Hurwitz	Constant fiber length, branch-value polynomial, ramification sum $2n - 2$	Monic division and resultants over K ; theorems 10.3 and 10.4 and proposition 10.5 .
Rolle at a scale	Exactly $k - r$ roots of $P^{(r)}$ in an occupied k -root ball ($1 \leq r \leq k$); branch directions; nearest-neighbour identity	Characteristic-zero differentiation of initial polynomials, over K ; theorems 11.2 and 11.6 and corollary 11.5 .
Hensel factorization	Unique coprime factors with support in S^* ; parameter version with no domain shrinking	Finite CRT linearization and well-founded recursion, over $\mathcal{O}_K$ and over $\mathcal{H}_{\Gamma, \geq 0}(U)$ for one fixed U ; theorems 12.1 and 12.5 .
Continuity of root multisets	Optimal ε/n matching, with multiplicity, no separation hypothesis	Finite balanced valuation-ball partition over $\mathcal{O}_K$; theorem 13.1 .
Simple-root perturbation	Rootwise threshold $T(P)$; full distance matrix and leading coefficients preserved; exact displacement; second-order Newton error	Product expansion plus valuation Rouché over $\mathcal{O}_K$; theorem 13.3 and corollaries 13.4 and 13.7 .
Residue and trace duality	Perfect residue pairing through collisions over an arbitrary commutative ring; trace determinant is the discriminant	Monic division, explicit dual basis, finite CRT, formal Laurent coefficients; theorems 14.1 , 14.2 and 14.6 . Overlap with the supplied results is credited.
Nullstellensatz and local multiplicity	Finite-system solvability with a class-safe certificate; finite local-algebra lengths	Classical algebra in each set-sized workspace $K[x]$; theorem 15.2 and proposition 15.4 .
Bézout, Bézoutian, Euler–Jacobi	Finite free rank $\prod_i d_i$ over an arbitrary commutative ring; coefficient-independent perfect pairing; Jacobian trace formula through collisions	Degree-decreasing rewriting with a critical-pair argument, divided-difference determinant, kernel map; theorems 16.1 , 17.1 and 17.2 and corollary 17.3 .
Comparison with a coherent contour residue	$\Lambda_F = \lambda_F$ in a delimited polynomial subclass	Degree bookkeeping plus errors of positive support, over $\mathcal{H}_{\Gamma, \geq 0}(U)$ for one fixed U ; theorem 17.5 . Conditional on both hypotheses; see section 19 .

20.5 Directions justified by the proved results

An effective representation of root-cluster refinements for restricted computable classes of surreal coefficients, with certified precision propagation for resultants and multiple critical clusters, would extend [section 13](#). Comparing support-controlled parameter factors with ramification and monodromy across the ordinary discriminant would extend [theorem 12.5](#). Replacing the coordinate-power highest terms of [\(16.1\)](#) by a homogeneous regular sequence with a known finite basis, using a graded

division and duality normalization in place of the rectangular argument, would extend [section 16](#); the existing general theory of complete intersections and Bézoutians is relevant, but a fixed-basis arbitrary-ring statement must specify its hypotheses rather than assume freeness. A perturbation theory for repeated-root trees would have to preserve a nested collection of cluster algebras with explicit precision thresholds, since a theorem preserving individual labels is impossible through genuine collisions. The general analytic residue comparison for the full class of positive-support holomorphic perturbations needs another argument. These are directions for additional work, not conclusions silently assumed in the present proofs.

A Finite-dependency and support audit

The only genuinely infinite operations in this article are coprime factor lifting and the coefficientwise expansions attached to it. All root-product identities, derivatives, matrices, residue calculations, normal forms and CRT decompositions involve only finitely many factors or coordinates. For lifting, an output exponent receives finitely many support-word contributions by [lemma 2.4](#); this remains true for a group of arbitrary rank and for well-ordered supports with infinitely many terms below some upper bound, which is exactly [remark 2.6](#).

Operation	Size or support control
Finite polynomial addition and multiplication	Finite unions and Minkowski sums of coefficient supports.
Monic division of a finite polynomial	Finitely many ring operations; integral coefficients remain integral.
Initial polynomial at (a, ρ)	One finite translation and dilation, one minimum, finitely many standard parts.
Newton profile and root-cluster tree	Finitely many active indices or root pair separations; no cofinal sequence of scales.
Coprime factor correction	All outputs lie in S^* ; at each exponent only earlier correction coefficients occur.
Parameter factor correction	The same S^* , with one holomorphic finite matrix inverse on the original fixed domain U .
Hölder matching	A finite partition of the combined root multiset; no choice of an infinite limiting matching.
Threshold stability estimate	Exact finite product expansion with explicit valuation remainders.
Multivariate normal form	At most $\deg H$ uses of an error on a branch; support in $S + \bigcup_{k \leq L} kT$ by (16.3) , with no positivity needed.
Residue, trace, Bézoutian	Finite quotient matrices or finitely many formal local Laurent coefficients; no contour limit.

B Notation index

Notation	Meaning
$\mathbf{SC} = \mathbf{No}[i]$	Surcomplex class field.
$t^\gamma = \omega^{-\gamma}$	Conway normal-form monomial.
$\Gamma, F = F_\Gamma, K = K_\Gamma$	Set-sized divisible exponent group; the real and complex Hahn workspaces $\mathbb{R}((t^\Gamma)), \mathbb{C}((t^\Gamma))$.
$ z , v(z), \text{lc}(z)$	Surreal-valued modulus; natural valuation; leading complex coefficient.
$\mathcal{O}_K, \mathbf{m}_K, \text{st}$	Finite elements, infinitesimals, and the standard part $\mathcal{O}_K \rightarrow \mathbb{C}$.
$D(a, r), \overline{D}(a, r)$	Open and closed order disks defined by the modulus.
$B_{\geq}(a, \rho), B_{>}(a, \rho)$	Closed and open valuation balls, $v(z - a) \geq \rho$ and $v(z - a) > \rho$.
$a + t^\rho(c + \mathbf{m}_K)$	Residue-direction ball at center a , scale ρ , direction $c \in \mathbb{C}$.
$w_{a,\rho}(P), I_{a,\rho}(P)$	Weighted Gauss valuation and the ordinary complex initial polynomial in the variable X .
j_-, j_+	Smallest and largest indices attaining the weighted coefficient minimum.
ϕ_P	Newton valuation profile at center zero.
S^*	Additive monoid generated by a positive well-ordered support, including zero.
$d_{ij}, d_i, \delta_i, T(P)$	$v(\alpha_i - \alpha_j); v(P'(\alpha_i)) = \sum_{j \neq i} d_{ij}; \max_{j \neq i} d_{ij}$; the threshold $\max_i(d_i + \delta_i)$.
E, ε	Perturbation and its coefficient precision $\min_j v([z^j]E)$.
$A_P = K[z]/(P), \lambda_P$	Finite quotient and its top-remainder coefficient functional $[z^{n-1}] \text{rem}_P$.
$m_h, \text{Res}(P, Q), \text{Disc}(P)$	Multiplication endomorphism; resultant; discriminant.
Δ_P, b^i	Univariate Bézoutian and the residue-dual basis (14.2).
$\mathcal{B}, D, \rho_\sharp, x^{\text{top}}$	Rectangular exponent set, its size $\prod_i d_i$, the top degree $\sum_i(d_i - 1)$, and the top monomial.
$\text{NF}_F, \lambda_F, \Delta_F, J_F$	Multivariate normal form, residue functional, Bézoutian, Jacobian determinant.
Λ_F	The supplied coherent contour residue (17.14), used only in theorem 17.5.
$\mathcal{H}_\Gamma(U), \mathcal{H}_{\Gamma, \geq 0}(U), \mathcal{H}_{\Gamma, > 0}(U)$	$\mathcal{O}(U)((t^\Gamma))$ on one fixed ordinary domain U , its nonnegative-support subring, and the positive-support ideal of that subring.

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